EPFL

WIRELESS RECEIVERS - ALGORITHMS AND ARCHITECTURES

OFDM Project Report

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GROUP 20

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1 TASK 1

1.1 SYSTEM DESIGN AND IMPLEMENTATION

In this section, we describe the acoustic OFDM transmission system implemented in MATLAB. The complete frame consists of an RRC-shaped single–carrier preamble followed by one known OFDM symbol used for channel estimation and 50 QPSK OFDM data symbols. All signals are ultimately transmitted as a real-valued audio waveform at $f_s = 48$ kHz over an $f_c = 8$ kHz acoustic carrier.

1.1.1 TRANSMITTER ARCHITECTURE

Figure 1 illustrates the transmitter processing chain.

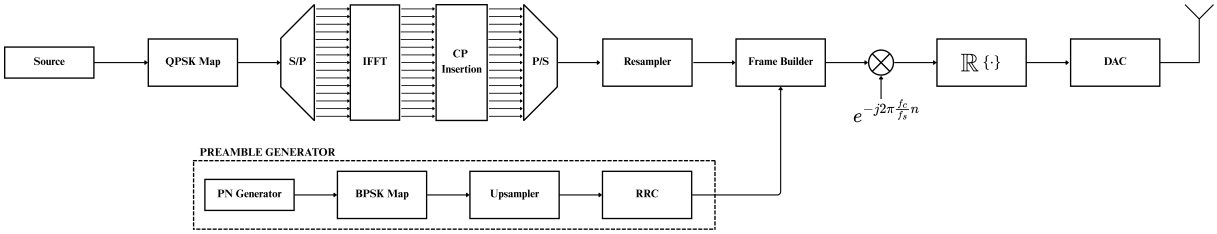


FIGURE 1
Transmission chain diagram of the implemented acoustic OFDM system.

SOURCE The source block generates the binary information bits to be transmitted. In our implementation this corresponds to a pseudo-random bit sequence of length $512 \times 2 \times (1 + 50)$, matching one known OFDM symbol and 50 data symbols.

QPSK MAP Bits are grouped into pairs and mapped to a normalized Gray-coded QPSK constellation. The output is a complex-valued sequence of unit-average-power symbols that serves as input to the OFDM modulator.

SERIAL-TO-PARALLEL (S/P) The QPSK symbol stream is reshaped into $N = 512$ parallel subcarrier streams and organized into an OFDM grid. The first column contains the known training symbol, while the remaining 50 columns carry data symbols. Each column thus represents one frequency-domain OFDM symbol.

IFFT For every OFDM symbol, a 512-point IFFT converts the frequency-domain grid column to its time-domain representation at the natural OFDM sampling rate $f_{\text{OFDM}} = 2$ kHz. This yields 512-sample baseband OFDM symbols.

CP INSERTION A cyclic prefix of length $N_{\text{CP}} = 256$ samples is prepended to each OFDM symbol by copying its last 256 samples. The CP ensures that the channel convolution appears circular and enables one-tap per-subcarrier equalization at the receiver.

PARALLEL-TO-SERIAL (P/S) The CP-extended symbols are concatenated into a single complex baseband OFDM waveform at $f_{\text{OFDM}} = 2$ kHz.

RESAMPLER The complex OFDM baseband is resampled from 2 kHz to the audio rate 48 kHz using the provided `ofdm_tx_resampling` function. This step includes appropriate anti-alias filtering and amplitude normalization so that the OFDM payload has unit average power and is compatible with the loudspeaker/audio interface.

PREAMBLE GENERATOR In parallel to the OFDM chain, a preamble generator produces a real-valued single–carrier preamble at 48 kHz. A binary PN sequence is generated by an LFSR, mapped to BPSK symbols, upsampled to 48 kHz and shaped with the provided RRC pulse. The resulting waveform is normalized to have the same average power as the OFDM payload. This preamble is later used for frame synchronization at the receiver.

FRAME BUILDER The frame builder concatenates the preamble and the resampled OFDM payload into a single complex baseband signal at 48 kHz. Internally, the preamble is treated as a real-valued sequence (imaginary part zero), while the OFDM part is complex.

DIGITAL UPCONVERSION AND DAC The complex baseband signal is upconverted to passband by multiplying with $e^{j2\pi f_c n/f_s}$, and the real part of the product is taken to obtain the audio waveform $s[n]$ centered at $f_c = 8$ kHz. Finally, $s[n]$ is scaled to the range $[-1, 1]$ and sent to the DAC and loudspeaker (or to the channel emulator) for acoustic transmission.

1.1.2 RECEIVER ARCHITECTURE

Figure 2 shows the receiver processing chain implemented in `rxofdm.m`.

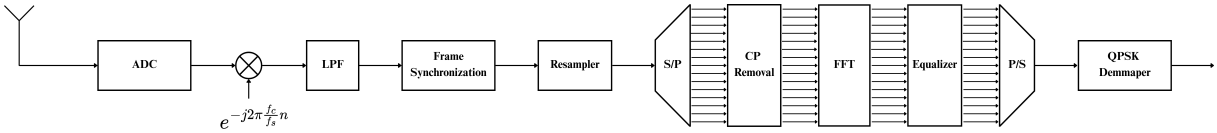


FIGURE 2
Receiver chain diagram of the implemented acoustic OFDM system.

ADC The real-valued acoustic signal is acquired either from the emulator output or from the microphone and sampled at $f_s = 48$ kHz. The resulting discrete-time sequence is the input to the digital receiver.

DOWNCONVERSION MIXER The sampled signal is mixed with $e^{-j2\pi f_c n/f_s}$ to shift the spectrum from passband back to complex baseband, recovering the in-phase and quadrature components of the transmitted signal.

LOW-PASS FILTER (LPF) A baseband low-pass filter removes out-of-band noise and residual images outside the OFDM band. The cutoff is chosen to pass the useful 2 kHz OFDM bandwidth while attenuating higher-frequency components produced by the digital mixing.

FRAME SYNCHRONIZATION The preamble portion of the baseband signal is processed by an RRC matched filter using the same pulse as in the transmitter. The matched-filter output is correlated with the locally regenerated PN–BPSK preamble, and the main correlation peak is detected automatically, using a dynamic detection threshold that adapts to channel conditions. The resulting index marks the start of the OFDM part of the frame.

RESAMPLER Starting from the detected frame index, the OFDM portion of the original complex baseband signal (before matched filtering) is extracted and resampled from 48 kHz back to $f_{\text{OFDM}} = 2$ kHz. This reverses the transmitter-side resampling and ensures that each OFDM symbol contains exactly $N_{\text{CP}} + N$ samples.

SERIAL-TO-PARALLEL (S/P) AND CP REMOVAL The resampled signal is segmented into 51 consecutive blocks of 768 samples (256-sample CP + 512-sample OFDM symbol) and converted to parallel form. For each block, the CP is discarded, leaving 512-sample OFDM symbols without prefix.

FFT A 512-point FFT is applied to each CP-free symbol to obtain the received frequency-domain subcarrier values $Z[m, s]$.

EQUALIZER The first received OFDM symbol corresponds to a known QPSK training symbol and is used to estimate the channel frequency response. Let $Z[m, 1]$ denote the FFT output of the first OFDM symbol at subcarrier index m, and let $A_{\text{known}}[m]$ be the locally regenerated QPSK symbol transmitted on that subcarrier. A per-subcarrier least-squares (LS) channel estimate is obtained as

$$H_{\text{long}}[m] = \frac{Z[m, 1]}{A_{\text{known}}[m]}$$

To enforce a physically meaningful channel length and reduce noise, the LS estimate is converted to the time domain via an IFFT to obtain the discrete-time channel impulse response $h_{\text{long}}[n]$. The effective channel length L_h is detected as the last tap whose magnitude exceeds 1% of the maximum tap amplitude. The impulse response is then truncated to $h_{\text{trunc}}[0{:}L_h - 1]$ and zero-padded back to length N, after which a refined frequency response is computed as

$$\hat{H}[m] = \text{FFT}\{h_{\text{trunc}}[n]\}$$

This truncated LS estimate provides a smooth and robust channel model that suppresses low-energy tails beyond the effective delay spread. For the remaining 50 OFDM data symbols, one-tap frequency-domain equalization is applied using $\hat{H}[m]$, optionally combined with phase-tracking schemes (Decision-Directed or Viterbi–Viterbi) to compensate for time-varying phase distortions. They are detailed in the channel-estimation section of the report.

PARALLEL-TO-SERIAL (P/S) AND QPSK DEMAPPER The equalized QPSK symbols are converted back to a serial stream and demapped to binary values using minimum-distance decoding with respect to the Gray mapping. The resulting bitstream is the estimate of the originally transmitted data.

1.2 VERIFICATION ON EMULATED CHANNEL ID 1

To validate the correct operation of the implemented OFDM system, we transmit one full frame through the emulated channel ID 1 at an SNR of 100 dB. This channel acts essentially as an AWGN channel with negligible distortion, and therefore serves as a baseline test to confirm that all transmitter and receiver blocks—including frame synchronization, OFDM demodulation, channel estimation, and equalization—are functioning correctly.

Constellation before and after equalization. Figure 3 shows the received QPSK symbols before and after frequency–domain equalization. Before equalization, the received symbols already form four tight clusters almost perfectly aligned with the ideal QPSK points. This is expected, since channel ID 1 introduces no phase distortion and the SNR is extremely high (100 dB), resulting in a very reduced symbol spread. After equalization, the minimal symbol spread completely disappears, confirming correct FFT processing and LS estimation in this ideal-channel scenario.

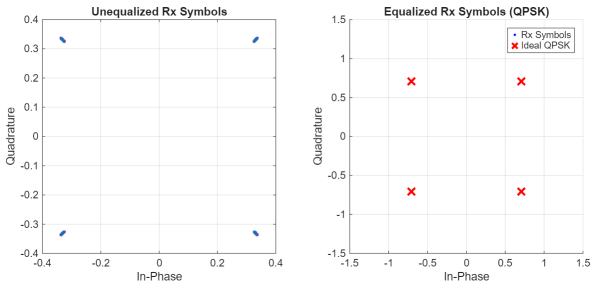


FIGURE 3
Unequalized (left) and equalized (right) received QPSK symbols for channel ID 1 at 100 dB SNR.

Estimated channel impulse response. Figure 4 shows the magnitude of the LS-estimated channel impulse response, obtained from the known OFDM symbol. The channel consists of a dominant single tap, with all remaining taps at noise level. This further confirms that channel ID 1 is effectively flat, and that the estimation process is stable and correctly implemented.

Frame synchronization. Figure 5 shows the normalized correlation metric used for frame synchronization, together with the RRC matched-filter output around the detected preamble peak. A clear, isolated maximum is obtained, and the detected boundary aligns precisely with the start of the OFDM payload. This verifies correct preamble detection and sample-accurate alignment of the OFDM symbols.

Conclusion. All tests confirm that the system performs correctly on the simplest emulated channel. The constellation,

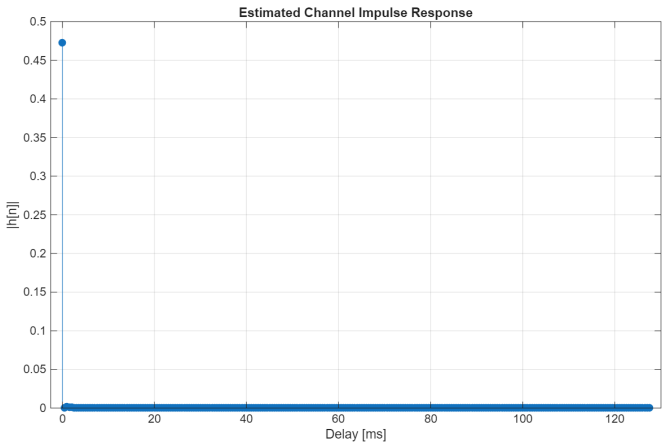


FIGURE 4
Estimated channel impulse response for channel ID 1. Only one significant tap is present, as expected.

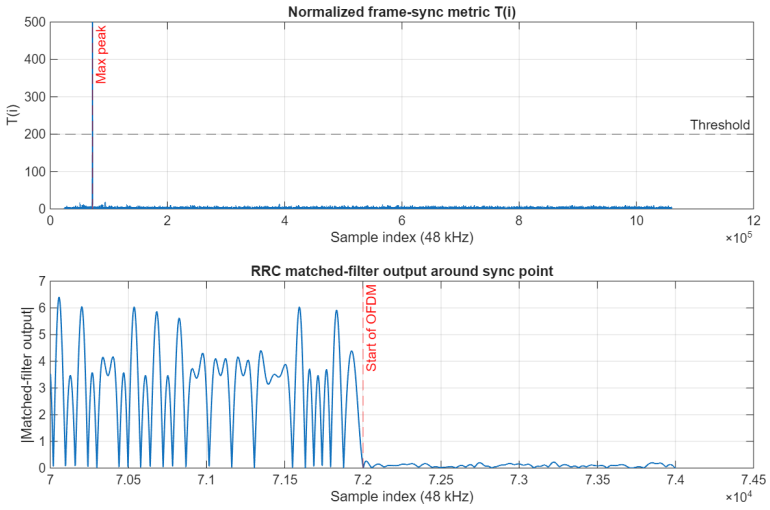


FIGURE 5
Frame synchronization results: normalized correlation metric (top) and RRC matched-filter output around the synchronization point (bottom).

impulse response, and synchronization results collectively verify that the entire transmitter–receiver chain is operating as expected under ideal channel conditions.

1.3 BER PERFORMANCE AS A FUNCTION OF SNR

To validate Task 1, the system BER is evaluated as a function of SNR using the default configuration and channel ID 1. For each SNR point, ten independent OFDM frames are transmitted and the BER is averaged. The resulting BER curve is shown in Fig. 6.

As expected, the BER decreases with increasing SNR and drops below $\mathrm{BER} < 1\%$ at approximately $\mathrm{SNR} \approx -5.5$ dB. For $\mathrm{SNR} \geq -5.5$ dB, the current setup supports reliable data transmission, whereas lower SNR regimes would require configuration changes—such as reduced throughput or added redundancy—to achieve comparable performance.

2 TASK 2: PHASE TRACKING

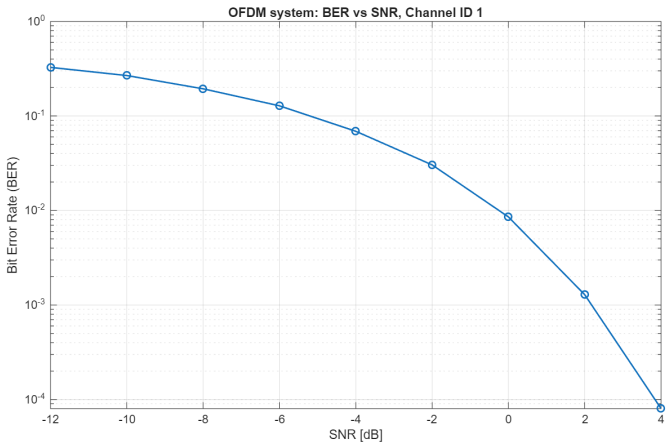


FIGURE 6
BER performance versus SNR for the OFDM system over channel ID 1.

2.1 TRANSMISSION OVER CHANNEL ID 2 WITHOUT PHASE TRACKING

Emulated channel ID 2 introduces an oscillator mismatch that manifests as a *symbol-wise* common phase error (CPE). Figure 7 shows that, for each OFDM symbol (colour), all subcarriers undergo the same phase rotation, while the rotation changes from symbol to symbol. Hence the impairment is not frequency-selective; it is a time-varying CPE common to all subcarriers.

After one-tap equalization with the (static) LS estimate from the known OFDM symbol, the equalized constellation collapses into a noisy ring rather than four QPSK clusters, because the equalizer cannot compensate the symbol-to-symbol phase drift. Accordingly, essentially no data symbols are decoded reliably and the measured BER is $\mathrm{BER} \approx 0.4705$, i.e., close to random decisions. A dedicated phase-tracking mechanism is therefore required for channel ID 2 to recover OFDM symbols beyond the first few.

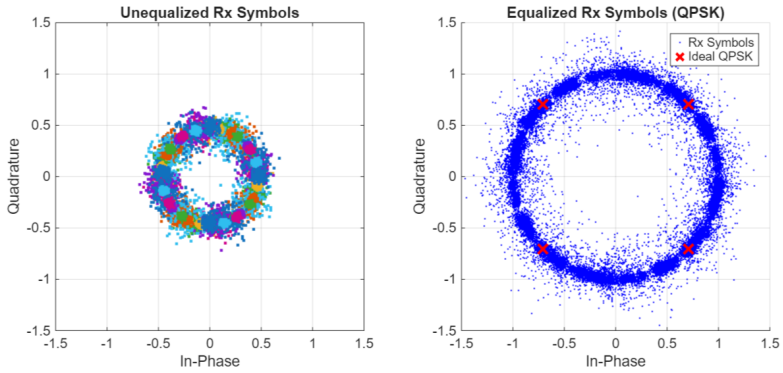


FIGURE 7
Unequalized (left) and equalized (right) received QPSK symbols over channel ID 2 without phase tracking. Each colour corresponds to one OFDM symbol.

2.2 GLOBAL CPE TRACKING

Since the phase impairment is common to all subcarriers, estimating a single phase per OFDM symbol is sufficient and more robust than per-subcarrier tracking.

2.2.1 Algorithm

Let $Y_n(k)$ denote the one-tap equalized symbol on subcarrier n at OFDM symbol index k. We project each subcarrier onto the unit circle and average across subcarriers to obtain a low-noise CPE observation, then apply optional exponential smoothing:

$$\widehat{\theta}(k) = (1 - \alpha_{\mathrm{CPE}})\,\widehat{\theta}(k-1) + \alpha_{\mathrm{CPE}}\,\arg\left(\frac{1}{N_{\mathrm{sc}}}\sum_{n=1}^{N_{\mathrm{sc}}}\frac{Y_n(k)}{|Y_n(k)|}\right).$$

The corrected symbols are obtained as $Y_n(k)\,e^{-j\widehat{\theta}(k)}$.

2.2.2 Tuning of α_{CPE}

We sweep $\alpha_{\mathrm{CPE}} \in [0, 1]$ and measure the BER for channel ID 2 at $\mathrm{SNR} = 100$ dB (Fig. 8). The best result is obtained for $\alpha_{\mathrm{CPE}} = 0$, confirming that the phase changes so quickly that temporal averaging is detrimental; the tracker must use the instantaneous per-symbol CPE estimate.

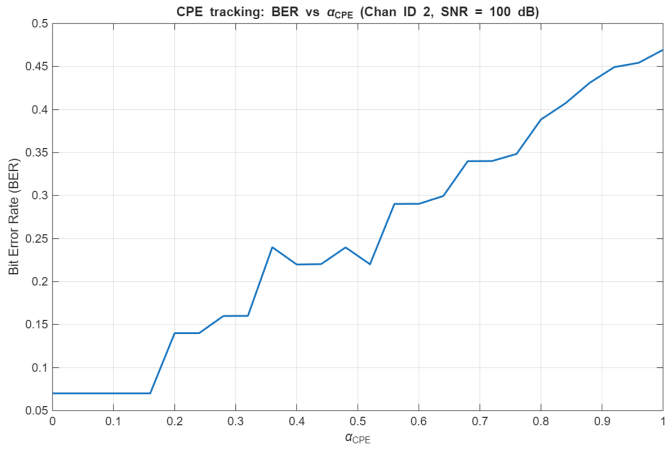


Figure 8
BER as a function of α_{CPE} for global CPE tracking in channel ID 2 at high SNR. The optimum is $\alpha_{\mathrm{CPE}} = 0$, i.e., no temporal smoothing.

2.2.3 Results on Channel ID 2

With $\alpha_{\mathrm{CPE}} = 0$, Fig. 9 shows compact QPSK clusters, indicating that the common phase rotation is effectively removed (until loss of lock). Figure 10 shows the estimated phase evolution: the accumulated phase grows approximately smoothly, while the symbol-to-symbol increment increases with symbol index. The tracker maintains accurate alignment until approximately symbol 44; beyond that point it loses lock abruptly. In terms of Task 2, this means that *without* tracking essentially no data OFDM symbols are received successfully, whereas *with* global CPE tracking roughly the first ~ 44 OFDM symbols are decoded reliably before failure.

2.3 Why Phase Tracking Fails After a Finite Number of Symbols

All decision-aided phase trackers ultimately fail here due to QPSK phase ambiguity. In channel ID 2, the symbol-to-symbol phase increment increases until it approaches the QPSK decision ambiguity threshold of $\pi/4$. Once $\Delta\theta \gtrsim \pi/4$, the received constellation no longer provides an unambiguous reference for the correct quadrant, so the phase estimate becomes unreliable and the tracker loses lock. This explains the abrupt breakdown observed in Fig. 10 and why only a finite number of OFDM symbols can be recovered even with an optimal CPE estimator.

2.4 Alternative Phase Tracking Methods and Comparison

For completeness, we briefly discuss two standard alternatives to global CPE tracking and justify why they are suboptimal for channel ID 2 but preferred for the audio channel.

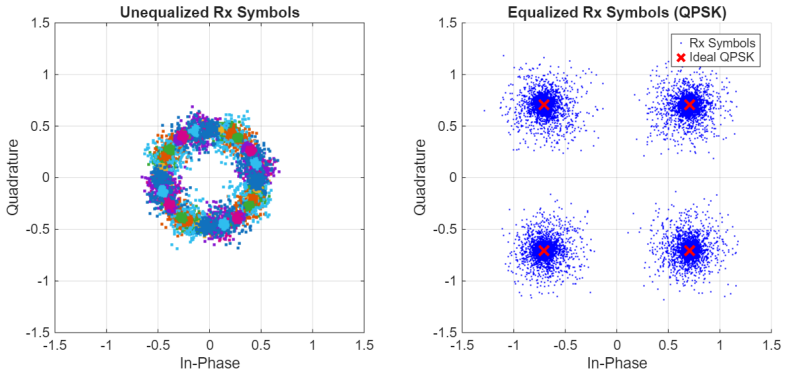


FIGURE 9
Equalized constellation under global CPE tracking ($\alpha_{\mathrm{CPE}} = 0$). The four QPSK clusters are substantially more compact than without tracking.

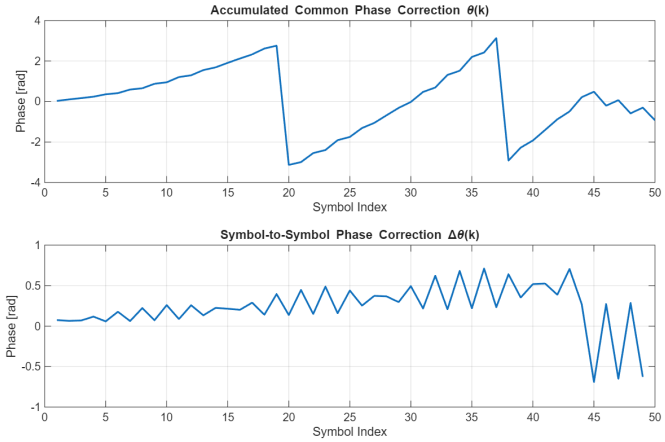


FIGURE 10
Accumulated phase correction (top) and symbol-to-symbol phase increments (bottom) using global CPE tracking in channel ID 2.

DECISION-DIRECTED (DD) TRACKING. Decision-directed tracking updates a per-subcarrier complex channel estimate using hard symbol decisions. For each OFDM symbol index s and subcarrier k, the channel estimate is recursively updated as

$$H_k(s) = \alpha_{\mathrm{DD}}\, H_k(s-1) + (1-\alpha_{\mathrm{DD}})\, \frac{Z_k(s)}{\hat{a}_k(s)},$$

where $Z_k(s)$ is the received frequency-domain symbol after FFT, $\hat{a}_k(s)$ is the hard QPSK decision, and $\alpha_{\mathrm{DD}} \in [0,1]$ controls the tracking memory. DD tracking is well suited to slowly varying multipath channels, where both amplitude and phase evolve gradually and may differ across subcarriers.

In channel ID 2, however, the impairment is dominated by a rapid symbol-wise common phase rotation. DD tracking therefore attempts to follow a non-existent frequency-selective distortion and becomes highly sensitive to decision errors. As shown in Fig. 11 (left), DD can only follow the phase drift during the first part of the frame; once the phase increment grows, the per-subcarrier estimates diverge and the tracker loses lock after roughly 37 symbols.

PER-SUBCARRIER VITERBI–VITERBI (VV) TRACKING. The Viterbi–Viterbi algorithm estimates phase by exploiting the rotational symmetry of PSK constellations. For each subcarrier k and OFDM symbol index s, the instantaneous phase estimate is

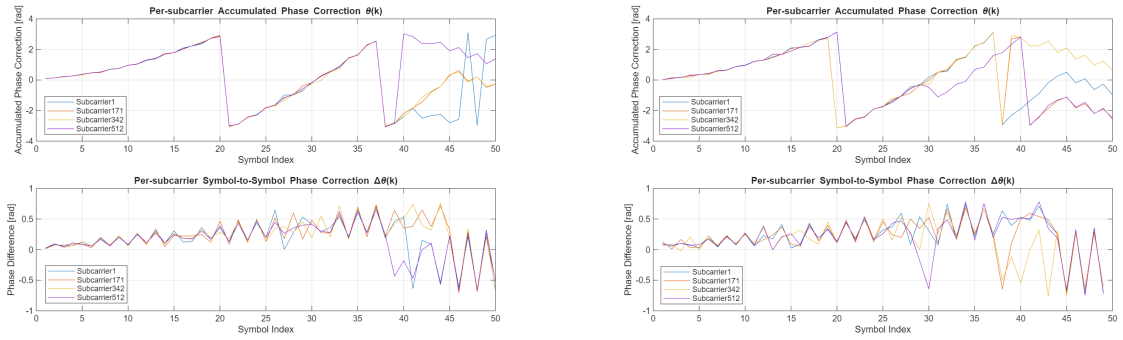
$$\Delta\theta_k(s) = \frac{1}{4}\angle\big(-(z_k(s)/H_k)^4\big)\,,$$

where $z_k(s)$ is the received FFT output and H_k is the static LS estimate obtained from the known OFDM symbol. The estimate is optionally smoothed via

$$\theta_k(s) = \theta_k(s-1) + \alpha_{\mathrm{VV}}\,\Delta\theta_k(s),$$

and applied as $\tilde{z}_k(s) = z_k(s)\,e^{-j\theta_k(s)}$.

Per-subcarrier VV tracking is effective when phase distortions vary across frequency, as in frequency-selective phase noise or multipath channels. In channel ID 2, however, the phase distortion is identical for all subcarriers. The VV phase trajectories in Fig. 11 (right) remain nearly identical while tracking succeeds and then break down simultaneously as the phase increment grows, confirming that per-subcarrier processing provides no additional information in this scenario.



DD tracking VV tracking

FIGURE 11
Per-subcarrier accumulated phase correction (top) and symbol-to-symbol phase increments (bottom) for DD (left) and VV (right) tracking in channel ID 2.

COMPARISON AND DISCUSSION. In channel ID 2, the phase distortion is a purely symbol-wise common phase error, so estimating a single phase per OFDM symbol maximizes noise averaging and avoids unnecessary per-subcarrier degrees of freedom. This allows global CPE tracking to maintain lock significantly longer than DD or VV and to achieve the lowest BER in this emulated scenario. In a real acoustic channel, however, residual Doppler, multipath, and time-varying propagation effects typically induce subcarrier-dependent phase and amplitude variations. Under such conditions, per-subcarrier methods such as DD or Viterbi–Viterbi tracking become preferable, as a single global phase estimate is no longer sufficient to capture the channel dynamics.

3 TASK 3: CHANNEL CHARACTERIZATION

In this section, we characterise the emulated channels ID 2 to ID 5 based on diagnostic plots obtained from the transmission of a fixed OFDM frame. For each channel, we analyse the unequalized and equalized constellations, frequency response (magnitude and phase), and time-domain representations (channel impulse response (CIR) and power dealy profile (PDP) when relevant), in order to identify the dominant propagation impairments. For a detailed explanation of how we calulated PDP refer to Appendix 8.1.

3.1 CHANNEL ID 2

Channel ID 2 was fully analysed in Task 2. Its defining characteristic is a *deterministic, time-varying common phase error* that accumulates across OFDM symbols due to oscillator mismatch. A global CPE estimator can track and compensate this phase drift only until the symbol-to-symbol increment exceeds the QPSK ambiguity threshold of $\pi/4$, at which point all decision-directed phase tracking methods lose lock. No other impairments (multipath, frequency selectivity, or noise enhancement) are present in this channel.

3.2 CHANNEL ID 3

Channel ID 3 exhibits the behaviour of a static, frequency-selective multipath channel. The unequalized constellation shows circular clouds around the QPSK points, indicating a fixed subcarrier-dependent complex gain with no time variation. After equalization, the symbols collapse exactly onto the QPSK grid, yielding zero BER.

The frequency response displays periodic ripples in both magnitude and phase, which are the classical signature of multipath interference. The CIR and PDP reveal three significant taps: a dominant main path and two weaker delayed echoes. All delays lie well within the cyclic prefix, preventing ISI and ICI and explaining the perfect equalization performance.

In summary, Channel 3 is a static multipath channel with limited delay spread fully contained within the CP, resulting in frequency selectivity but no time variation or phase instability.

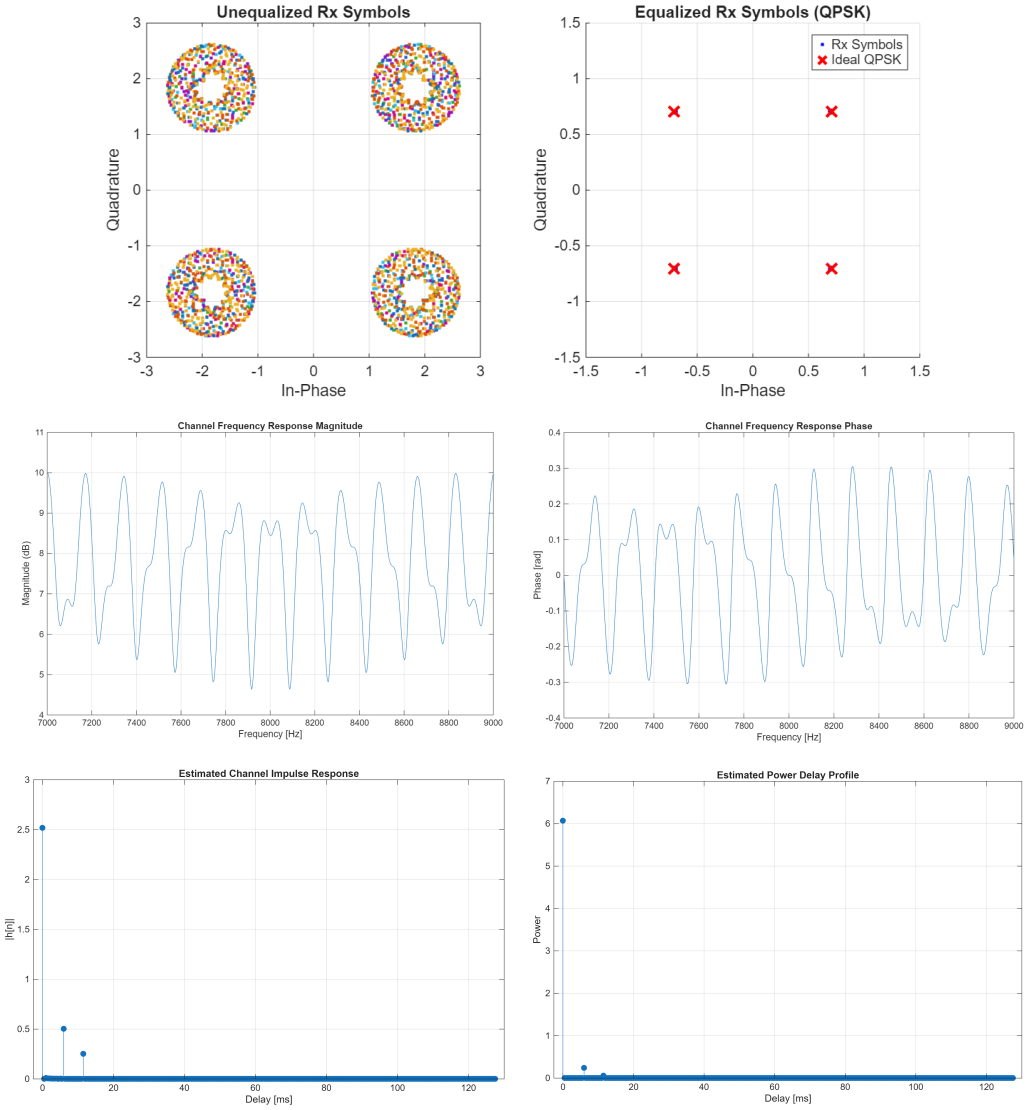


FIGURE 12
Channel 3: unequalized and equalized constellations (top), frequency response magnitude and phase (middle), CIR and PDP (bottom).

3.3 CHANNEL ID 4

Channel ID 4 is a strongly band-limited single-path channel. The unequalized constellation collapses along straight lines, reflecting severe frequency-dependent attenuation that causes large groups of subcarriers to align along common directions. After equalization, only the central subcarriers form clean QPSK clusters, while edge subcarriers remain noise-dominated due to excessive noise amplification.

The frequency response magnitude confirms a narrow passband centred within the OFDM band, with attenuation below -35dB at the edges. The phase response is nearly linear within the passband, indicating a stable single-path channel with constant group delay. The PDP shows a single dominant tap with negligible delay spread, fully contained within the CP.

In summary, Channel 4 is not multipath-limited but spectrally limited: its high BER is caused by severe attenuation of edge subcarriers rather than dispersive or time-varying effects.

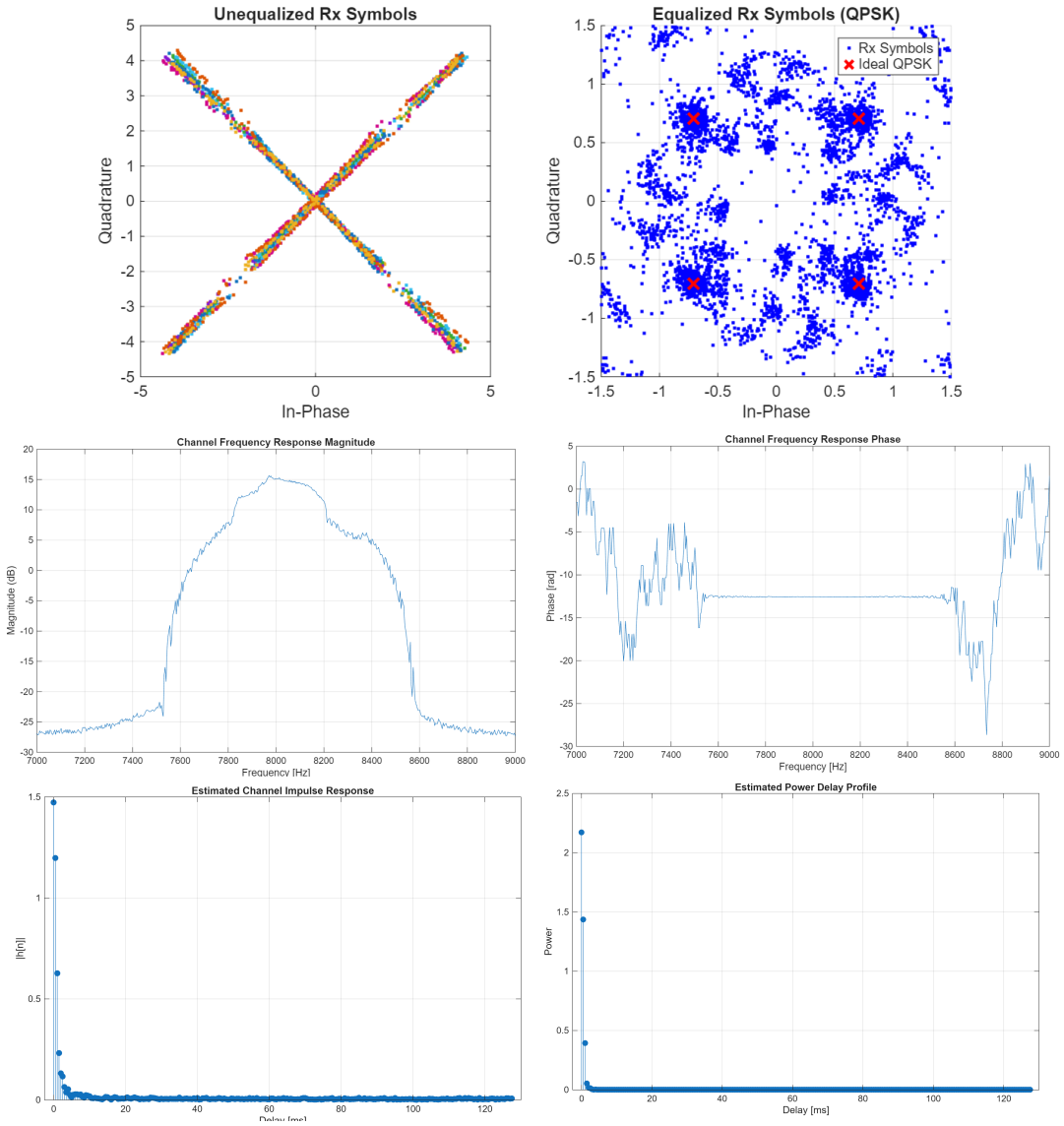


FIGURE 13
Channel 4: unequalized and equalized constellations, frequency response magnitude and phase, CIR and PDP.

3.4 CHANNEL ID 5

Channel ID 5 behaves as a static, deterministic LTI channel with a frequency response shaped by short-delay multipath propagation. The unequalized constellation exhibits ring and swirl patterns caused by subcarrier-dependent amplitude scaling and linear phase rotation, which remain perfectly stable over time.

After equalization, the constellation collapses into tight QPSK clusters with zero BER, confirming the absence of time variation, common phase error, or stochastic impairments. The channel frequency response magnitude displays a smooth U-shaped profile without deep fades, while the phase response is nearly linear, indicating an approximately constant group delay. Both effects are consistent with constructive and destructive interference among a small number of closely spaced multipath components.

The estimated impulse response and PDP show a dominant main tap with rapidly decaying secondary taps, confirming the presence of short-delay multipath. However, the delay spread is sufficiently small to induce no observable ISI, rendering the channel effectively single-tap from an OFDM system perspective.

In summary, Channel 5 is a static, deterministic, and easily invertible channel whose smooth magnitude shaping and near-linear phase response are direct consequences of short-delay multipath, with no time selectivity or phase instability.

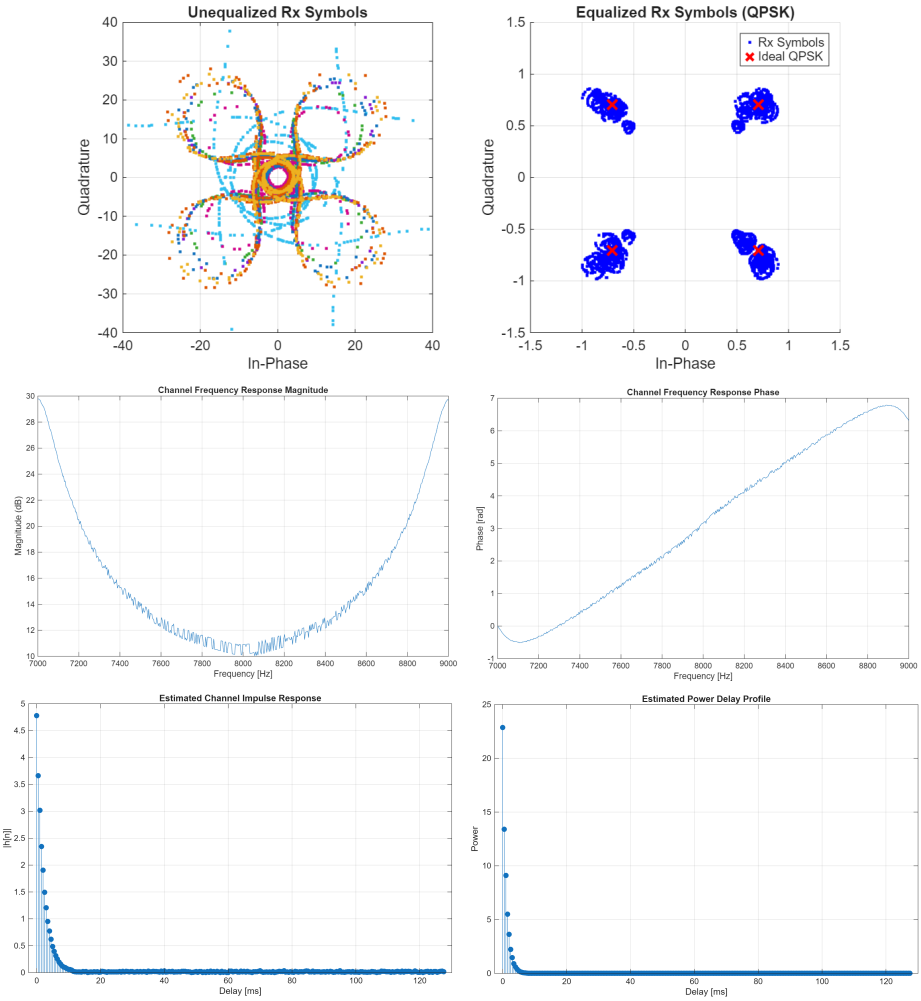


FIGURE 14
Channel 5: unequalized and equalized constellations, frequency response magnitude and phase, CIR and PDP.

4 TASK 4

4.1 SPECTRAL EFFICIENCY OF OUR OFDM SYSTEM

Spectral efficiency quantifies how efficiently an OFDM system uses the occupied bandwidth to transmit information. It is defined as

$$\eta = \frac{R}{B} \quad \text{[bit/s/Hz]},$$

where R is the achievable bit rate (bit/s) and B is the occupied bandwidth (Hz).

OCCUPIED BANDWIDTH

In OFDM, N subcarriers are placed with uniform spacing Δf. Thus, $B = N\Delta f$.

USEFUL BITS PER SECOND

We use QPSK, which carries $M = 2$ bits per subcarrier. Therefore, we have $N \cdot M$ bits/subcarrier, and Δf subcarrier/sec. Thus we have

$$N \cdot M \cdot \Delta f \quad \text{bits/s}$$

CYCLIC PREFIX EFFICIENCY

Each OFDM symbol also contains a cyclic prefix of length N_{CP} symbols. Only N subcarriers carry new information, leading to the CP efficiency

$$\eta_{\text{CP}} = \frac{N}{N + N_{\text{CP}}}.$$

DATA OFDM SYMBOLS EFFICIENCY

Each transmitted frame contains N_{TS} training OFDM symbols and N_S data OFDM symbols. Thus,

$$\eta_{\text{TS}} = \frac{N_S}{N_S + N_{\text{TS}}}.$$

SPECTRAL EFFICIENCY COMPUTATION

The achievable bit rate is thus $R = \eta_{\text{TS}} \cdot \eta_{\text{CP}} \cdot (NM\Delta f)$ bits/s

Since $B = N\Delta f$, the spectral efficiency becomes

$$\eta = \frac{R}{B} = \eta_{\text{TS}} \cdot \eta_{\text{CP}} \cdot M.$$

$$\boxed{\eta = \left(\frac{N_S}{N_S + N_{\text{TS}}}\right)\left(\frac{N}{N + N_{\text{CP}}}\right) M}$$

NUMERICAL VALUES

We now substitute the parameters of our system:

$$N = 512, \quad N_{\text{CP}} = 256, \quad N_S = 50, \quad N_{\text{TS}} = 1, \quad M = 2.$$

Therefore,

$$\eta = \left(\frac{50}{51}\right)\left(\frac{512}{512 + 256}\right)(2) = \left(\frac{50}{51}\right)\left(\frac{2}{3}\right)(2) \approx 1.31 \text{ bit/s/Hz}.$$

$$\boxed{\eta \approx 1.31 \text{ bit/s/Hz}}$$

PARAMETERS AFFECTING SPECTRAL EFFICIENCY

From the final expression, we see that the spectral efficiency depends on the following system parameters:

- M : number of bits per QAM/PSK symbol (modulation order),
- N_S : number of data OFDM symbols per frame,
- N_{TS} : number of training OFDM symbols per frame,
- N : number of subcarriers per OFDM symbol,
- N_{CP} : cyclic prefix length.

It *does not* depend on the speaker sampling rate (48 kHz) (and thus oversampling factor).

Note that for a fixed bandwidth B, varying the subcarriers spacing Δf would vary the total number N of subcarriers, thus varying the spectral efficiency.

4.2 METHODS TO IMPROVE SPECTRAL EFFICIENCY AND THEIR LIMITATIONS

From the final expression, the spectral efficiency can be increased by improving any of the three factors: Data OFDM symbol efficiency, CP efficiency, or modulation order. Below we discuss how each parameter can be modified and the corresponding limitations.

1. **Increase N (number of subcarriers)**

   For fixed CP length, the ratio $\frac{N}{N+N_{\mathrm{CP}}}$ increases with N, so CP overhead becomes negligible as $N \to \infty$, and $\eta_{CP} \to 1$.

   Limitations:

   - Increasing the number N of subcarriers decreases the inter-subcarrier spacing $\Delta f = B/N$. This increases sensitivity to any frequency offset when recovering the received symbols, because any frequency offset would lead to stronger ICI.
   - A larger FFT size increases latency and computational complexity.

2. **Decrease the cyclic prefix length N_{CP}**

   A smaller CP directly increases $\eta_{\mathrm{CP}} = N/(N + N_{\mathrm{CP}})$ for a fixed N.

   Limitations:

   - N_{CP} has a lower bound: it must exceed the effective delay spread (length) of the channel. If N_{CP} is smaller than the channel length, the last samples of the previous subcarrier start interfering with the current subcarrier, and we thus observe increased ICI.
   - In practice, we can go around this lower bound by obtaining the channel PDP, idenifying the noise floor, and choosing N_{CP} as the length for which the channel power is greater than this noise floor.

3. **Increase modulation order M**

   Using higher-order constellations (16-QAM, 64-QAM, etc.) increases the number of bits per subcarrier and thus increases η linearly.

   Limitations:

   - For a higher order constellation, higher SNR is required to achieve a same BER performance.

4. **Reduce training overhead or increase the number of data OFDM symbols N_S**

   Spectral efficiency increases when the fraction of useful OFDM symbols

   $$\eta_{\mathrm{TS}} = \frac{N_S}{N_S + N_{\mathrm{TS}}}$$

   gets larger. This can be achieved either by

- using fewer training OFDM symbols N_{TS}, or
- increasing the number of transmitted data OFDM symbols N_S per frame.

In our case, we only use 1 training OFDM symbol, so we only have the option of increasing the number of data OFDM symbols.
Limitations:

- Using fewer training for a same number of data symbols can weaken the initial channel estimate, specially for noisy channels or channels with a small coherence bandwidth, making equalization and phase tracking less accurate.
- Increasing N_S for a same number of training symbols makes the frame longer: if the channel phase varies significantly over time, long frames without additional pilots cannot track changes accurately in the channel, degrading BER.

5 TASK 5

5.1 CHARACTERISTICS OF THE REAL CHANNEL

For this task, we transmit a real sound wave using the provided speaker. The wave propagates through the physical acoustic channel and is captured by the microphone.

- The experiment is conducted in an extremely quiet room.
- The two speakers are placed facing each other with a separation distance of 30 cm.
- The microphone is positioned midway between the two speakers.
- The laptop volume is set to its maximum value to ensure consistent and comparable measurements.
- The number of transmitted data symbols is reduced from 50 to 10 in order to reduce the acquisition time.

To characterize the real audio channel, we perform a recording, compute a least-squares estimate of the channel as described in the Appendix, and analyze its frequency response, impulse response, and power delay profile (Figures (A)–(E)). Together, these plots provide a comprehensive description of how the physical environment shapes the channel behavior.

FREQUENCY-DOMAIN CHARACTERISTICS

Figures (A) and (B) show the magnitude and phase of the estimated channel frequency response over the transmission band. The magnitude response exhibits significant variations across frequency, with several pronounced notches and peaks spanning more than 20 dB. This behavior indicates a *frequency-selective channel*. The deep fades observed in the spectrum are characteristic of destructive interference caused by multipath propagation, where reflected components arrive with delays comparable to the inverse bandwidth and interfere with the direct path.

The phase response is irregular and non-linear across frequency, which indicates the presence of multipath propagation. Multiple paths with different delays combine at the receiver, producing a frequency-dependent phase response that cannot be described by a single delay.

Strong phase variations appear near frequencies where deep magnitude notches occur. At these frequencies, destructive interference causes the channel magnitude to become very small, so the channel response lies close to the origin of the complex plane. In this region, even small changes in frequency can lead to large changes in phase. These variations are continuous and reflect steep phase slopes rather than true phase discontinuities.

TIME-DOMAIN CHARACTERISTICS: IMPULSE RESPONSE

The estimated impulse response shown in Figure (C) reveals a dominant early tap corresponding to the direct line-of-sight (LOS) propagation between the speaker and the microphone. Following this main component, several additional taps with smaller but non-negligible amplitudes appear within the first few milliseconds. These taps represent early reflections from nearby surfaces such as the table, floor, or walls.

Beyond the first few milliseconds, the impulse response exhibits a long tail of very low-amplitude components extending over a large delay range. The absence of structured peaks at large delays and the extremely small magnitude of these components indicate that they should be interpreted as noise rather than as meaningful late echoes.

POWER DELAY PROFILE ANALYSIS

The power delay profiles displayed in Figures (D) and (E) is the same PDP graph but zoomed-in on the first few taps) support the interpretation of the impulse response graph. A dominant peak at very small delay indicates that most of the received energy arrives through the earliest propagation path (the line-of-sight path). Several weaker components follow shortly after, corresponding to early reflections that contribute only a small fraction of the total power. Beyond these initial delays, the power decays rapidly and remains at a very low level, showing that later arrivals carry negligible energy. This concentration of energy within a narrow delay window indicates that the channel has a short effective delay spread.

OVERALL INTERPRETATION

Overall, the measured acoustic channel is **LOS-dominant with noticeable multipath**. While the direct path carries most of the energy, early reflections are sufficiently strong to induce pronounced frequency selectivity, as evidenced by the deep fades in the frequency response. The effective delay spread remains short, implying limited inter-symbol interference provided an appropriate cyclic prefix is used.

Consequently, OFDM is well suited for this channel configuration. The observed frequency selectivity can be effectively handled through subcarrier-wise equalization in the frequency domain, allowing deep spectral fades to be compensated. Moreover, since the channel energy is confined to a short delay window, a relatively short cyclic prefix is sufficient to maintain subcarrier orthogonality, resulting in efficient and reliable transmission.

5.2 BER OBTAINED WITH THE REAL CHANNEL

With the same setup as the one in section 5.1, the BER obtained without phase tracking is 0.

This result is consistent with the channel characteristics identified in Section 5.1: although the channel is frequency selective due to multipath propagation, it is dominated by a strong LOS component and exhibits a short effective impulse response. These properties limit inter-symbol interference and allow the least-squares (LS) channel estimation and equalization alone to accurately recover the transmitted signal.

As expected, when we introduce any of the phase tracking methods, the BER obtained stays 0.

5.3 SPECTRAL EFFICIENCY IMPROVEMENT USING THE REAL CHANNEL

According the the channel's PDP which was taken by recording an average of the channel impulse responses and squaring, we can observe dominant taps with significant amplitudes until a delay of 2.5 ms. The rest of the taps are too small in amplitude and we can rule them out as representative of noise power. Therefore, the channel delay span can be approximated as 2.5 ms.

However, since the PDP results from an averaging process and may slightly underestimate rare but non-negligible delay components, a conservative margin is introduced to ensure robustness, where we extend the assumed delay span to 6 ms. This corresponds to number of samples equal to $6 \text{ ms} \cdot f_s = 6 \text{ ms} \cdot 2kHz = 12$ samples.

Therefore, that gives us the possibility of confidently reducing the cycle prefix length N_{CP} down to 12 samples, while still observing no inter-carrier-interference.

This would greatly benefit the spectral efficiency, becoming:

$$\eta = \left(\frac{50}{51}\right)\left(\frac{512}{512+12}\right)(2) = \left(\frac{50}{51}\right)\left(\frac{512}{524}\right)(2) \approx 1.916 \text{ bits/s/Hz}$$

which is very close to the ideal spectral efficiency of 2 bits/s/Hz (without any overhead).

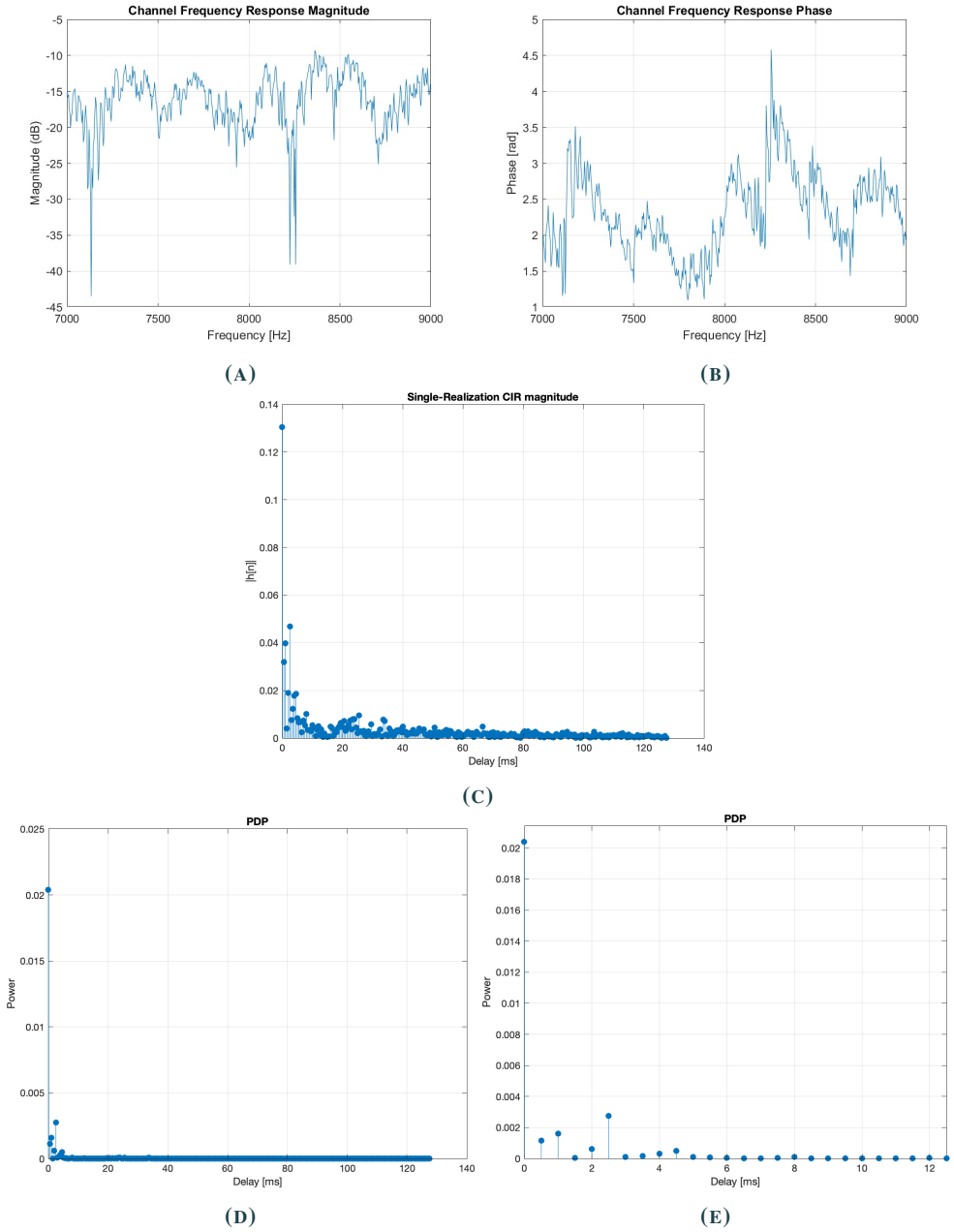


FIGURE 15
Channel frequency and time response for the audio case

6 ADDITIONAL WORK

Even though we have already included some additional plots and comments in the previous tasks, in this section we introduce further theoretical concepts related to our system and transmission, as well as some new experiments.

6.1 MOVING-MICROPHONE EXPERIMENT

In this part of the project, we investigated the robustness of our OFDM system when the receiver microphone is moving. This scenario is considerably more challenging than the previous static configurations because the acoustic channel changes rapidly as the microphone position varies, introducing time-varying delay, amplitude fluctuations and Doppler-like effects. The physical setup photo for this experiment is provided in Appendix 8.4 (Fig. 28d).

6.1.1 PERFORMANCE OF THE ORIGINAL RECEIVER

Using the original receiver architecture from the first part of the report—with only one training symbol at the beginning of the frame—we observed that the system performance degraded severely under even moderate microphone motion. For most motion trajectories, the measured BER was close to $50\,\%$ (i.e., essentially random decoding), indicating that the channel estimate was no longer valid for subsequent OFDM symbols. Only when the microphone remained very close to its initial position, moving only slightly in front of the loudspeakers, did we achieve acceptable performance with a BER below $25\,\%$.

This behaviour is expected: the original system assumes that the channel remains approximately constant over the entire frame and that it can be represented by a single channel estimate. When the microphone moves, even by a few centimeters, the multipath structure changes significantly, producing a new effective impulse response, but more importantly shifting the phase of the constellations all over the place. As a result, the channel estimation used in the first tasks becomes invalid almost immediately.

6.1.2 INCREASING THE TRAINING-SYMBOL DENSITY

Instead of changing the channel estimator itself, we tried to improve robustness by increasing the number of training symbols. Concretely, we adopted a periodic scheme: one known OFDM training symbol is transmitted every K data OFDM symbols. The receiver updates its channel estimate whenever a new training symbol is received and uses this most recent estimate in the tracking calculations described in the first part of the report for the following K data symbols.

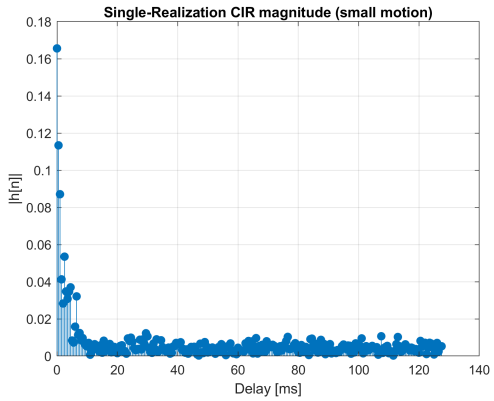
We repeated the moving-microphone experiment with this modified frame structure. The microphone was hand-held and moved along two different trajectories in front of the loudspeakers:

- *Small motion:* the microphone was moved slowly left and right by only a few centimetres in front of the loudspeakers, without changing its distance significantly.
- *Large motion:* the microphone was moved more aggressively, both sideways and in distance (tens of centimetres), so that the direct path and multipath pattern changed substantially over the duration of the frame. Additionally, the rotation of the constellations in their space is much more present (to be discussed in the following part).

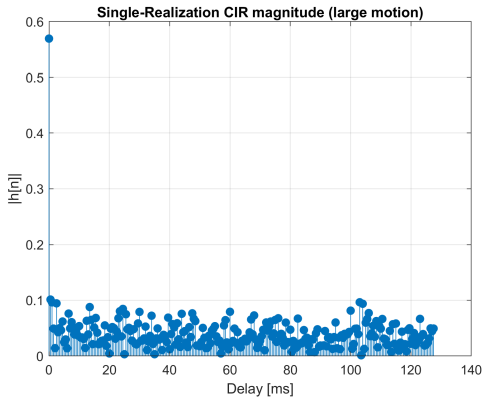
For the small-motion trajectory, the system was able to track the channel reasonably well with the periodic training symbols: the measured BER was around $\mathrm{BER}_{\mathrm{move,small}} \approx 5\,\%$, clearly worse than the static LOS case but still far from random. In contrast, for the large-motion trajectory the performance deteriorated dramatically despite the increased training density, with a BER of approximately $\mathrm{BER}_{\mathrm{move,large}} \approx 40\,\%$. In this regime the channel changes so quickly that even frequent training cannot provide a sufficiently accurate, up-to-date estimate for the data symbols, and the effective channel seen by the receiver is strongly time-varying within each OFDM symbol.

These results highlight that periodically inserting more training symbols can improve robustness to moderate motion, but it is not sufficient when the acoustic channel evolves rapidly and the microphone is moving significantly. In such cases, more sophisticated equalization strategies that explicitly model the time-dispersive and time-varying nature of the channel would be required.

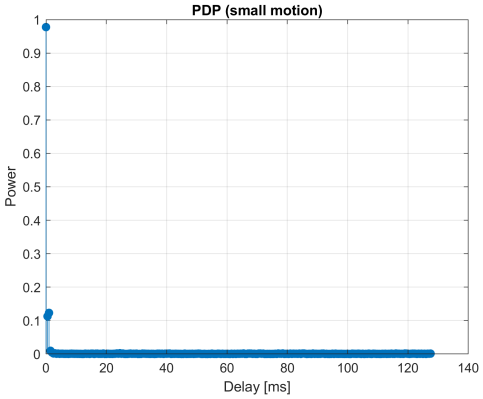
These BER results are consistent with the underlying channel behaviour observed from the estimated channel impulse responses and power–delay profiles (PDPs) in Figure 16. In the small-motion case, the PDP remains relatively compact, with a dominant early tap and only a few weaker reflections whose relative power varies slowly over time. In contrast, for large motion the channel impulse response becomes more spread and irregular, and the PDP exhibits additional delayed taps and small ripples after the first prominent paths, whose relative strength changes rapidly. This increased delay spread and time selectivity makes it difficult for a simple per-symbol channel estimate to remain valid over several OFDM symbols.



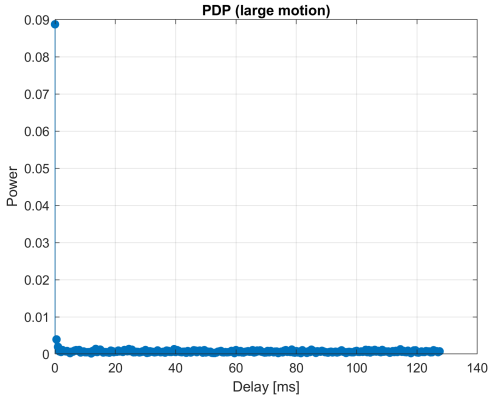
(A) CIR for small microphone motion.



(B) CIR for large microphone motion.



(C) PDP for small microphone motion.



(D) PDP for large microphone motion.

FIGURE 16

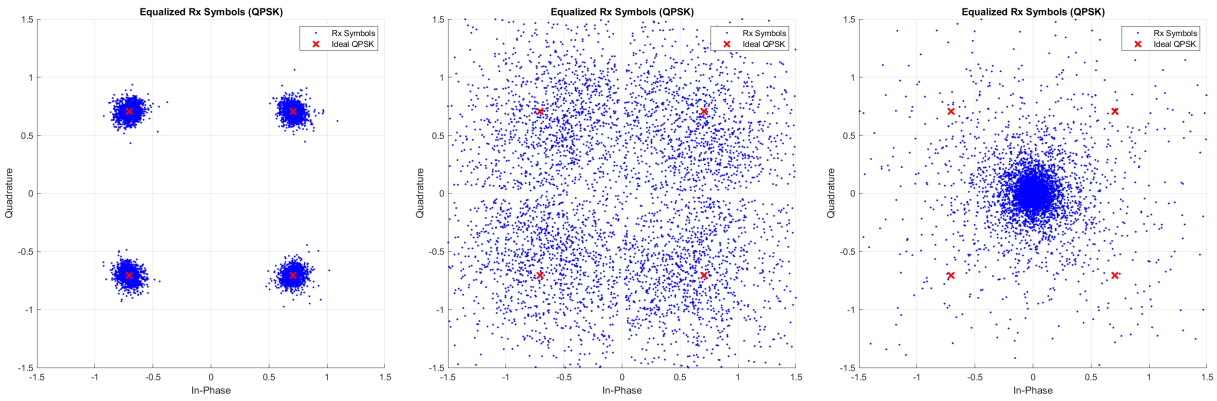
Estimated channel impulse responses (top row) and power–delay profiles (bottom row) for small and large microphone motion (first 256 taps). The increased delay spread and rapid path-power fluctuations in the large-motion case are consistent with the higher BER observed in the moving-microphone experiment.

EFFECT OF MICROPHONE MOTION ON THE EQUALIZED CONSTELLATION. To further visualize the impact of microphone motion, we compared the distribution of the equalized QPSK constellation points in three scenarios: (i) static microphone, (ii) small motion, and (iii) large motion. The results are shown in Figure 17. In the static case, the equalized symbols cluster tightly around the ideal QPSK points, indicating that the channel estimate and phase correction remain valid throughout the frame. With slow motion, the clusters become noticeably broader and slightly distorted, which is consistent with a slowly time-varying channel and residual phase/amplitude tracking errors. For large motion, the constellation becomes strongly dispersed and partially collapses toward the origin, showing that the equalization/tracking is no longer able to compensate the rapidly changing acoustic channel.

A key reason why even small movements can severely affect performance is the carrier-phase sensitivity: moving the microphone by approximately one wavelength produces a full 2π phase cycle. At our carrier frequency $f_c = 8\,\mathrm{kHz}$, the acoustic wavelength is $\lambda \approx c/f_c \approx 343/8000 \approx 4.3\,\mathrm{cm}$, so displacements on the order of only a few centimetres can already induce strong constellation rotation; in a multipath environment, the simultaneous change of several paths makes the distortion even more pronounced.

Before equalization, the received QPSK symbols on each subcarrier are multiplied by the (generally complex) channel frequency response and by any residual global phase rotation. As a result, each transmitted constellation point is rotated (phase) and scaled (magnitude) by an amount that depends on the subcarrier and can also vary with the OFDM symbol index. When we plot all subcarriers and many OFDM symbols together, these different phase rotations accumulate into ring-like or cloud-like patterns rather than four tight QPSK clusters. When the microphone moves, even small displacements induce significant phase changes (on the order of 2π per wavelength), and in a multipath environment several paths change simultaneously. This makes the unequalized points sweep around and

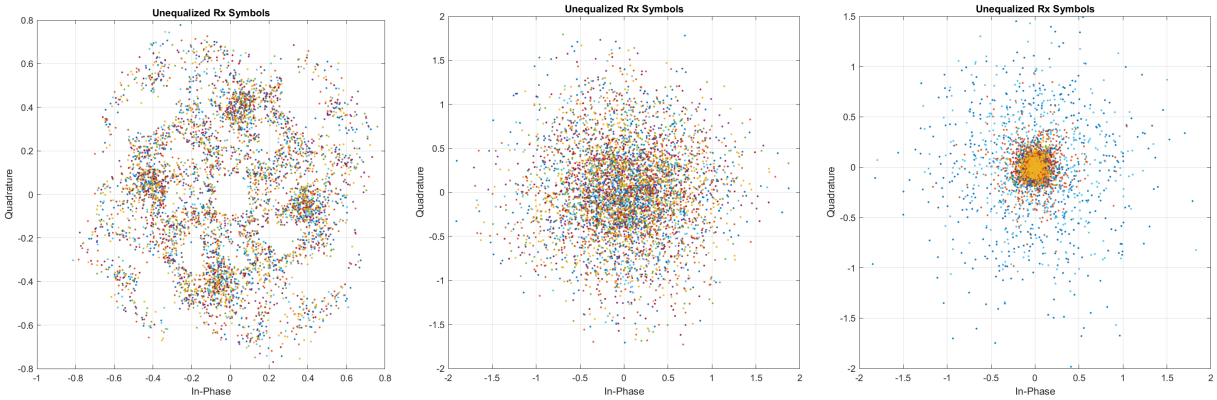
smear out further over time, which is why the slow/large-motion cases appear progressively more dispersed and irregular.



(A) Static microphone. (B) Small microphone motion. (C) Large microphone motion.

FIGURE 17

Equalized QPSK constellations for three microphone-motion regimes. Static reception yields tight clusters around the ideal points, slow motion increases the spread due to residual tracking errors, while large motion causes severe dispersion/collapse, indicating that the channel varies too rapidly for the receiver to track reliably.



(A) Static microphone. (B) Small microphone motion. (C) Large microphone motion.

FIGURE 18

Unequalized received constellations (symbol-colored points). Without equalization, unknown channel-induced phase shifts rotate the QPSK points differently across subcarriers and over OFDM symbols, producing ring/cloud-like patterns instead of four clusters. Increased motion leads to stronger time-varying phase (and amplitude) distortions, further smearing the constellation.

6.2 INFLUENCE OF EFFICIENCY ON BER – CYCLIC PREFIX LENGTH AND FREQUENCY OF TRAINING SYMBOLS

In our system the spectral efficiency is determined by the modulation order M, the cyclic prefix efficiency η_{CP} and the training symbol efficiency η_{TS}. Using the notation introduced in section 4.1, the overall efficiency can be written as

$$\eta = \eta_{\mathrm{TS}} \cdot \eta_{\mathrm{CP}} \cdot M,$$

where $\eta_{\mathrm{CP}} = \frac{N}{N+N_{\mathrm{CP}}}$ with N the number of useful samples per OFDM symbol, and $\eta_{\mathrm{TS}} = \frac{N_s}{N_s+N_{\mathrm{TS}}}$ accounts for the overhead of the inserted training symbols.

To study the trade-off between spectral efficiency and reliability, we first varied the cyclic prefix length N_{CP} while keeping all other parameters fixed (number of subcarriers, modulation, training frequency and carrier frequency). In

this experiment, the microphone was slowly moved left and right by a few centimetres in front of the loudspeakers, at an approximate distance of 5–10 cm from the speaker, so that the channel experienced moderate time variations. Figure 19 shows the measured BER as a function of N_{CP}. For very short cyclic prefixes, e.g. $N_{\mathrm{CP}} = 64$, the CP might be shorter than the effective channel impulse response, so inter-symbol interference is not completely removed and the BER remains relatively high. As we increase the CP to $N_{\mathrm{CP}} = 192$ and $N_{\mathrm{CP}} = 256$ samples, the BER decreases. At the same time, the cyclic-prefix efficiency η_{CP} decreases, so the overall efficiency drops. Beyond $N_{\mathrm{CP}} \approx 256$ samples the BER no longer improves, which suggests that the dominant channel taps are contained within this duration and that further increasing the CP only reduces efficiency without noticeable BER gain. However, as the BER values are in general very low in this scenario (on the order of 0.1%), even a shorter CP could work well which coincides with the fact that PDP has only dominant taps at low delays.

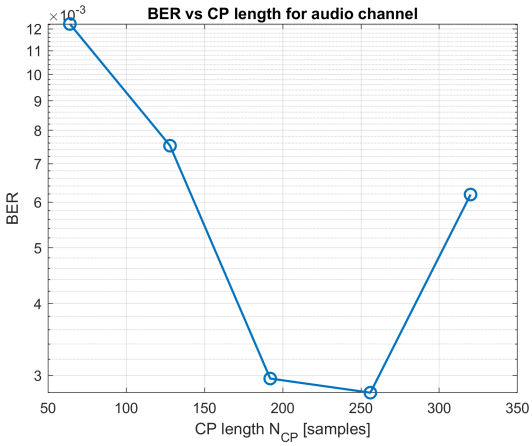


FIGURE 19
BER as a function of the cyclic prefix length N_{CP}.

In a second experiment, we fixed the cyclic prefix length to $N_{\mathrm{CP}} = 256$ (chosen as a good trade-off from the previous study) and modified the way training symbols are inserted compared to the first tasks. Instead of using the original training structure, we adopted a periodic scheme in which one training OFDM symbol is transmitted every K data OFDM symbols. In other words, the frame consists of repeating blocks of length $K + 1$, containing one training symbol followed by K data symbols, and the channel estimate at the receiver is updated whenever a new training symbol is received. The setup was the same as in the previous experiment.

In general, if a frame contains N_s data symbols and N_{TS} training symbols, the training-symbol efficiency is

$$\eta_{\mathrm{TS}} = \frac{N_s}{N_s + N_{\mathrm{TS}}}.$$

In our configuration, we transmit one training symbol every K data symbols, so asymptotically $N_s : N_{\mathrm{TS}} = K : 1$ and thus

$$\eta_{\mathrm{TS}} = \frac{K}{K+1}.$$

Larger K (and thus larger η_{TS}) increases the effective efficiency at the cost of less frequent channel updates. Figure 20 reports the BER as a function of overall efficiency η; each point corresponds to a different value of K through the relation above. For small K values (low η), e.g. $K = 1$, the channel is estimated very often, the system tracks phase and amplitude variations reliably, and the BER is low, at the cost of a relatively low efficiency. Increasing the spacing between training symbols improves the spectral efficiency, but the BER grows because the channel estimation becomes outdated in the presence of time variations and noise. This shows that the channel in our audio setup is not sufficiently static to allow very sparse training. Overall, these measurements highlight the expected trade-off: higher efficiency (larger η_{TS}) comes at the price of a higher BER, and the optimal working point depends on the acceptable error rate for the application. The points on the plot are the mean of 3 trials.

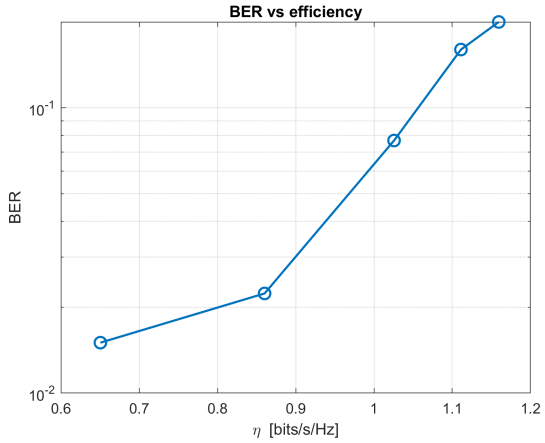


FIGURE 20
BER as a function of the efficiency η. Larger η (larger K) increases efficiency but also degrades the BER due to outdated channel estimates.

6.3 IMAGE TRANSMISSION OVER THE AUDIO OFDM LINK

To evaluate the end-to-end performance on a more perceptual task, we transmitted grayscale images through the complete audio OFDM chain. The images were first converted to a bitstream, and then passed to the transmitter as input using the same parameters as in the first tasks (number of subcarriers, cyclic prefix length and training frequency). At the receiver, the recovered bitstream was reassembled into an image for visual inspection. For each configuration we report both the resulting bit error rate (BER) and the peak-to-average power ratio (PAPR) of the transmitted OFDM frame. For detailes on PAPR refer to Appendix 8.2. The physical setup photos for the LOS and obstacle image-transmission scenarios are provided in Appendix 8.4 (Fig. 28b and Fig. 28c).

LINE-OF-SIGHT (LOS) CONFIGURATION. In the first scenario, the loudspeakers and the microphone were placed on two opposite sides of a table, at a distance of approximately $1\,\mathrm{m}$ with a clear line-of-sight (LOS) path between them. Under these conditions the channel is dominated by a strong direct component and a few weaker reflections, leading to relatively mild frequency selectivity.

The reconstructed image, shown in the centre panel of Figure 21, is visually close to the original: only a small amount of random pixel noise is visible. The measured bit error rate in this configuration is $\mathrm{BER}_{\mathrm{LOS}} \approx 0\,\%$, which is consistent with the BER values obtained for random data at the same spectral-efficiency setting. The PAPR of the transmitted OFDM frame is $\mathrm{PAPR}_{\mathrm{LOS}} = 28.09\,\mathrm{dB}$.

OBSTACLE CONFIGURATION. In the second scenario we introduced an obstacle between the loudspeakers and the microphone, while keeping the distance and system parameters unchanged. This attenuates the direct LOS path and increases the influence of multipath components, creating a more challenging acoustic channel.

The reconstructed image (right panel of Figure 21) exhibits clearly visible artifacts such as salt-and-pepper noise. The BER in this non-line-of-sight configuration increases to $\mathrm{BER}_{\mathrm{OBS}} \approx 5.24\,\%$, confirming that blocking the direct path significantly degrades system performance. Since the transmitted waveform is identical to the LOS case, the PAPR remains the same, with $\mathrm{PAPR}_{\mathrm{OBS}} = 28.03\,\mathrm{dB}$.



(A) Original (B) LOS reconstruction (C) Obstacle reconstruction

FIGURE 21
Original image (left) and reconstructed images after transmission over the audio OFDM link with LOS (centre) and with an obstacle between loudspeakers and microphone (right). The degradation in the obstacle case reflects the increased BER.

IMPACT OF BER AND PAPR ON VISUAL QUALITY. We observed that relatively small increases in BER (from $\mathrm{BER}_{\mathrm{LOS}} \approx 0\,\%$ to $\mathrm{BER}_{\mathrm{OBS}} \approx 5.24\,\%$) already have a substantial visual impact, as illustrated in Figure 21. As a result, even moderate BER values may corrupt a large fraction of the image. At the same time, the measured PAPR values of $28.03\,\mathrm{dB}$ indicate that the OFDM waveform contains high peaks that, in principle, can stress the loudspeaker and DAC if insufficient power backoff is used. In our audio setup, however, these PAPR levels did not cause any noticeable problems: the received images were reconstructed without visible clipping artefacts, indicating that the analogue front-end operated largely in its linear region for the chosen playback level.

Nevertheless, from OFDM theory and practical hardware design it is well known that large PAPR can lead to distortion and performance degradation when the power amplifier or loudspeaker is driven close to saturation. Therefore, acceptable performance for real data transmission should be evaluated jointly in terms of both BER and PAPR: a configuration with slightly higher BER but significantly lower PAPR may be preferable in hardware-limited scenarios, while in simulations the emphasis is often placed solely on BER.

PAPR REDUCTION TECHNIQUES FOR THE IMAGE-BASED OFDM WAVEFORM. High PAPR is a well-known limitation of OFDM systems, since large peaks in the time-domain signal can push the power amplifier, DAC, or loudspeaker into nonlinear operation. To quantify this effect for our image-transmission setup without performing additional over-the-air experiments, we performed an offline PAPR study directly on the baseband OFDM waveform generated from the image bit sequence. Concretely, we used the same OFDM parameters as in the previous experiments, mapped the image bits to QPSK symbols, formed OFDM frames, and computed the PAPR of the resulting time-domain signal. No channel or noise was simulated, and we therefore focus solely on PAPR (without reporting BER or visual quality for these variants).

We compared three configurations: (i) a baseline OFDM modulation of the image bitstream, (ii) frequency-domain interleaving of QPSK symbols across subcarriers, and (iii) whitening of the bitstream prior to QPSK mapping. Both interleaving and whitening act on the symbol/bit ordering only and do not change the efficiency.

- **Interleaving:** In the interleaving technique, the QPSK-modulated symbols associated with the image are permuted across the active subcarriers using a fixed pseudo-random permutation known at both transmitter and receiver. The symbol set remains unchanged, but modifying their ordering alters the peak structure of the time-domain OFDM signal by decorrelating neighbouring high-energy symbols, thus reducing the probability of constructive peak addition. As the permutation is random, we are not guaranteed to get a much better PAPR (if better at all). However, in principle, by searching over different permutations one could select the one that achieves the lowest PAPR.

  In the offline PAPR experiment on the image bitstream, interleaving reduced the PAPR from $28.03\,\mathrm{dB}$ in the baseline case to $27.88\,\mathrm{dB}$ (see Table 1).

- **Whitening:** In the whitening approach, the image bitstream is XORed with a pseudo-random binary sequence before QPSK mapping. This operation randomises the symbol statistics and reduces the formation of structured constellation patterns that tend to generate high time-domain peaks. The scrambling sequence is generated from a known seed so that the receiver can invert the operation perfectly.

  Whitening produced a stronger PAPR reduction than interleaving, lowering the PAPR from $28.03\,\mathrm{dB}$ for the baseline sequence to $12.98\,\mathrm{dB}$ for the whitened sequence.

These results show that even simple manipulations of the image bitstream (interleaving with some search for an advantageous permutation or whitening) can reduce the PAPR of the corresponding OFDM waveform. In a practical audio-OFDM link, such PAPR control can provide additional robustness against loudspeaker and DAC nonlinearities, especially when operating close to the hardware limits.

TABLE 1
Offline PAPR evaluation on the OFDM waveform generated from the image bitstream.

Method	PAPR [dB]
Baseline (image bits, no processing)	**28.03**
Interleaving	**27.88**
Whitening	**12.98**

Beyond the offline PAPR evaluation, we also tested whitening in the full end-to-end transmission chain. Using the same XOR-based scrambler with a fixed seed at the transmitter and the corresponding de-whitening at the receiver, we observed a clear qualitative improvement in the reconstructed image. In particular, the few erroneous pixels visible in the upper reconstructed-image example (without whitening) were no longer present when whitening was enabled, consistent with the reduced sensitivity to nonlinear distortion.

This observation was further supported by listening tests during the audio-mode transmission. Without whitening, the playback/recording chain occasionally produced audible "tac-tac-tac" artifacts indicative of saturation/clipping events. With whitening enabled, these saturation sounds were no longer audible, suggesting that the reduced PAPR kept the loudspeaker/DAC in a more linear operating regime. In Figure 22, we show the input image and the reconstructed outputs obtained with our audio-OFDM link. The left panel shows the original input image used for transmission. The middle panel shows the reconstructed output without whitening, where a few isolated pixel errors are visible. The right panel shows the reconstructed output with XOR-based whitening and corresponding de-whitening at the receiver; the previously observed pixel artifacts disappear, yielding a visually correct reconstruction.



(A) Original (B) No whitening (C) With whitening

FIGURE 22
Input image and reconstructed outputs without and with whitening.

6.4 ACOUSTIC OFDM CHANNEL CHARACTERIZATION

6.4.1 Measurement scenarios (channel profiles)

To evaluate the audio OFDM link under representative acoustic propagation conditions, we recorded transmissions in the following repeatable setups:

- **LOS (static):** speaker and microphone facing each other with a clear direct path.
- **Near (static):** same as LOS but with reduced distance.
- **OBS (static):** an obstacle blocks the direct path, so reception is dominated by reflections.
- **Wall reflections (static):** a strong reflector is used to create a pronounced echo path.
- **Corner (static):** speaker and microphone close to a corner, producing multiple strong reflections.
- **Moving:** microphone is moved during the frame, producing a time-varying channel.



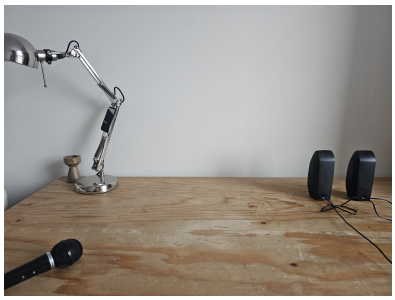
(A) LOS



(B) Near



(C) OBS



(D) Wall reflections



(E) Corner



(F) Moving

Figure 23
Photos of the physical setups used for the channel-measurement scenarios in Section 6.4.

6.4.2 Channel estimation and diagnostics

From the known training OFDM symbol we obtain the frequency response estimate

$$\hat{H}_0(f) = \frac{Z_{\text{train}}(f)}{A_{\text{train}}(f)}, \tag{1}$$

where $Z_{\text{train}}(f)$ denotes the received training subcarriers and $A_{\text{train}}(f)$ the known transmitted QPSK symbols. Over the subsequent OFDM data symbols we form an effective per-symbol estimate $\hat{H}(f, s)$ (depending on the chosen tracking mode) and visualize:

- **Frequency selectivity:** $20 \log_{10} |\hat{H}_0(f)|$ (deep notches indicate fades caused by multipath interference).
- **Time variation (normalized magnitude drift):**

$$20 \log_{10} \left| \frac{\hat{H}(f, s)}{\hat{H}(f, 1)} \right|, \tag{2}$$

  where values near 0 dB indicate a time-invariant channel.

- **Delay spread (PDP):** using $\hat{h}(\tau, s) = \mathrm{IFFT}\{\hat{H}(f, s)\}$ we plot $10 \log_{10} |\hat{h}(\tau, s)|^2$ over the first N_{CP} taps.
- **Equalization quality:** equalized QPSK constellation and $\mathrm{EVM}_{\mathrm{rms}}$ (lower is better). EVM quantifies residual distortion after equalization; lower is better; values >100% indicate severe mismatch. For a definition and interpretation of EVM, we refer to Appendix 8.5.

6.4.3 RESULTS AND DISCUSSION

SUMMARY VIA EVM. Table 2 summarizes the observed $\mathrm{EVM}_{\mathrm{rms}}$ across scenarios. LOS (static) yields the lowest EVM, while motion leads to severe degradation due to channel time variation. Multipath-heavy static setups (OBS / wall / corner) remain mostly time-invariant but exhibit stronger frequency selectivity and delay spread, increasing EVM due to noise enhancement in deep fades and residual mismatch.

TABLE 2
Equalization performance across acoustic scenarios (from constellation EVM).

Scenario	$\mathrm{EVM}_{\mathrm{rms}}$
LOS	6.20%
Near	11.95%
OBS	22.20%
Wall	21.09%
Corner	43.03%
Moving	105.42%

LOS (STATIC). Figures 24–26 show that the LOS channel is quasi time-invariant, with a smooth magnitude response $|\hat{H}_0(f)|$, negligible normalized drift (values close to 0 dB across frequency), and a PDP whose energy is concentrated at the earliest delays. This indicates a dominant direct path with only weak early reflections. Consequently, one-tap equalization is highly effective and the equalized constellation forms tight clusters around the ideal QPSK points, resulting in the lowest observed EVM.

NEAR. Despite the higher received power, the near configuration exhibits stronger frequency selectivity than LOS, visible as a bowl-shaped magnitude response with more pronounced notches in $|\hat{H}_0(f)|$. This behaviour is consistent with stronger early reflections that partially cancel the direct path at specific frequencies. Although the channel remains largely time-invariant, these deep fades amplify noise during equalization, explaining the higher EVM compared to the LOS case.

OBS AND WALL REFLECTIONS. When the direct path is blocked (OBS) or a strong reflector is introduced (wall), the PDP reveals increased delayed energy due to additional echo components, while $|\hat{H}_0(f)|$ contains deeper and narrower fades. In the obstacle-reflection case, the delayed echo produces a comb-like frequency selectivity with quasi-periodic notches. Although these channels remain mostly static over the frame, the stronger multipath interference leads to increased noise enhancement on deeply faded subcarriers and residual equalization mismatch, resulting in significantly higher EVM.

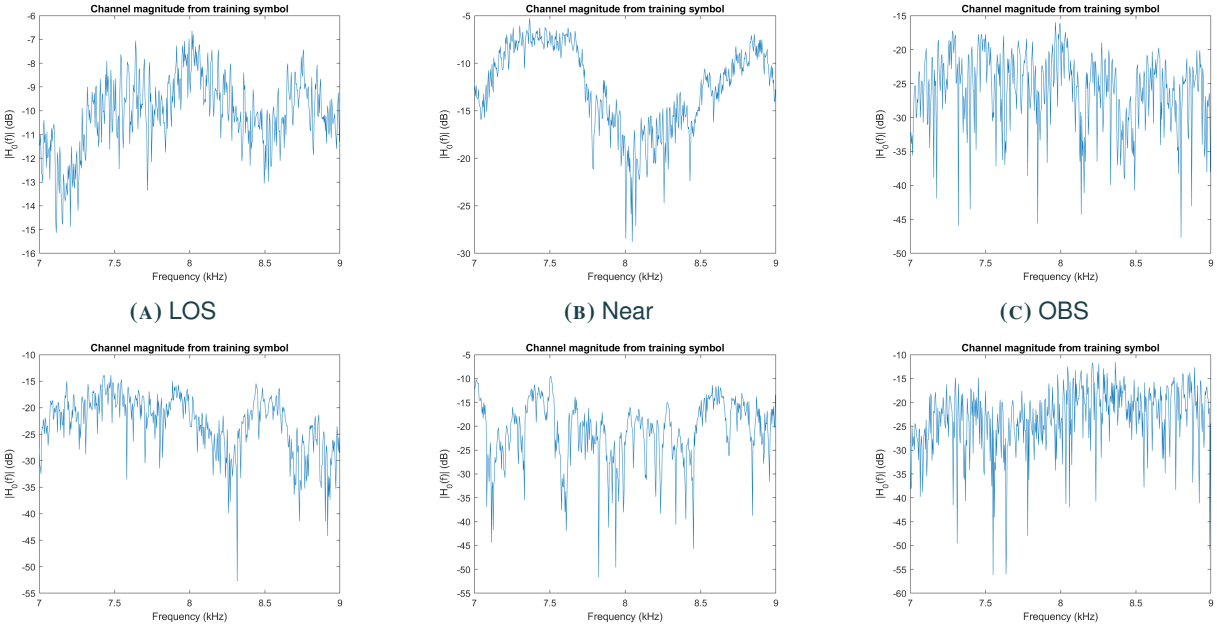
CORNER. Near a corner, multiple reflections with comparable strength arrive at the receiver, creating a highly frequency-selective channel with many deep notches and more structured drift patterns. In this regime, even small perturbations of the environment can noticeably alter the interference pattern across frequency. This combination of strong multipath and sensitivity to small changes yields the worst equalization performance among the static scenarios, with an EVM of approximately 43%.

MOVING. The moving scenario exhibits the strongest normalized magnitude drift across both time and frequency, indicating that the initial estimate $\hat{H}_0(f)$ becomes outdated during the frame. Together with the pronounced per-subcarrier phase drift (Appendix 8.6), this time variation invalidates the static equalizer, causing the equalized constellation to collapse into a diffuse cloud (EVM $> 100\%$). This behaviour confirms that the channel coherence time is comparable to or smaller than the transmission duration, motivating the use of more frequent pilot symbols or fully decision-directed tracking to maintain reliable demodulation under motion.

Complementary to the normalized magnitude drift in Fig. 26, we also inspected the per-subcarrier phase drift maps (Appendix 8.6), which confirm negligible phase evolution in static scenarios and increased, frequency-dependent phase fluctuations under motion and strong multipath.

NOTE ON APPARENT DRIFT IN STATIC SCENARIOS. Although the LOS/near/OBS/wall/corner setups are static, some panels in Fig. 26 still show narrow "drift stripes" at specific frequencies. This does not necessarily indicate a true time variation of the acoustic channel. Our drift metric is based on the ratio $\hat{H}(f, s)/\hat{H}(f, 1)$; when the channel magnitude $|\hat{H}(f, 1)|$ is very small (deep fades), the estimate has low effective SNR and the ratio becomes numerically unstable, which can create large-looking phase/amplitude variations even in a static channel. Consistently, these apparent drift stripes tend to occur at frequencies where Fig. 24 exhibits strong notches (deep attenuation), i.e., where equalization is most sensitive to estimation noise. Additionally, drift may appear due to the incorrect equalization as decission-dircted tracking amplifies the noise if it makes the incorrect decision.

NOTE ON AXIS SCALING AND VISUAL COMPARISON. In the figures presented in this section, some plots (in particular the channel magnitude responses and PDP visualizations) use axis limits or color scales that are optimized per scenario rather than being strictly identical across all panels. This choice was made to preserve the visibility of relevant structure within each scenario, since enforcing a common scale would compress several plots and obscure weaker but physically meaningful multipath components. As a consequence, direct visual comparison of absolute magnitudes across different scenarios should be interpreted with care. The quantitative comparisons in this section are therefore based on scale-invariant metrics (EVM and normalized drift) and on qualitative structural features (e.g., delay spread, presence of deep fades, and temporal variability), rather than on absolute axis values alone.

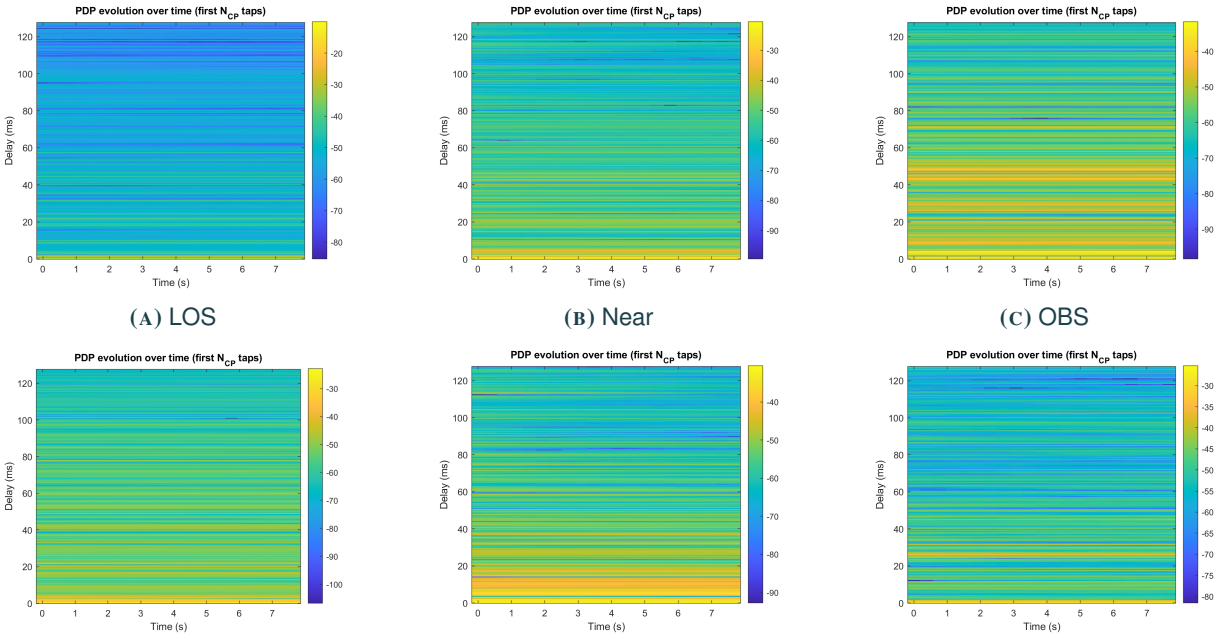


(A) LOS (B) Near (C) OBS

(D) Wall reflections (E) Corner (F) Moving

FIGURE 24

Channel magnitude from the training symbol. We plot $20 \log_{10} |\hat{H}_0(f)|$ versus frequency around the carrier. Deep notches indicate strong frequency-selective fading caused by multipath interference (destructive superposition of paths). LOS is comparatively smooth, while OBS/wall/corner exhibit deeper fades; the moving case additionally suffers from time variation (see Fig. 26).

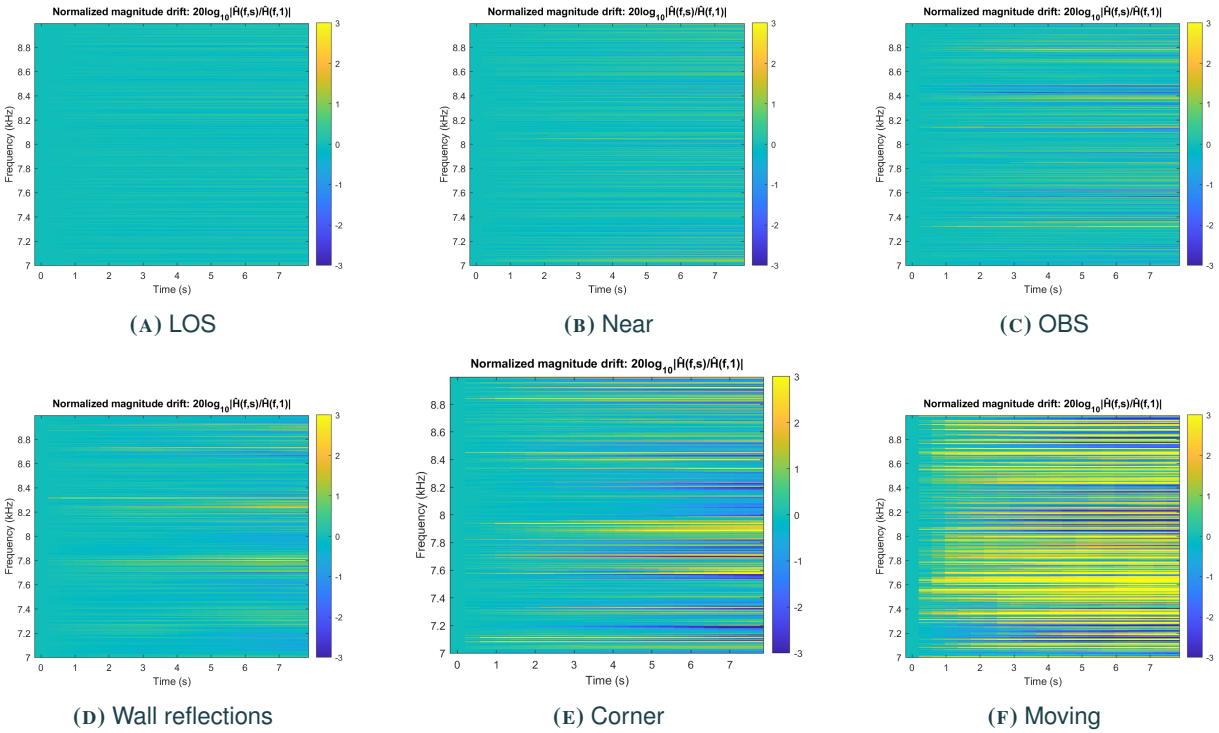


(A) LOS (B) Near (C) OBS

(D) Wall reflections (E) Corner (F) Moving

FIGURE 25

PDP evolution over time (first N_{CP} taps). Heatmaps show $10\log_{10}|\hat{h}(\tau,s)|^2$ where $\hat{h}(\tau,s)=\mathrm{IFFT}\{\hat{H}(f,s)\}$. Strong horizontal bands at non-zero delays correspond to stable echoes/reflections; broader spread of energy indicates increased delay spread. Multipath-rich setups (OBS/wall/corner) show more delayed energy than LOS.

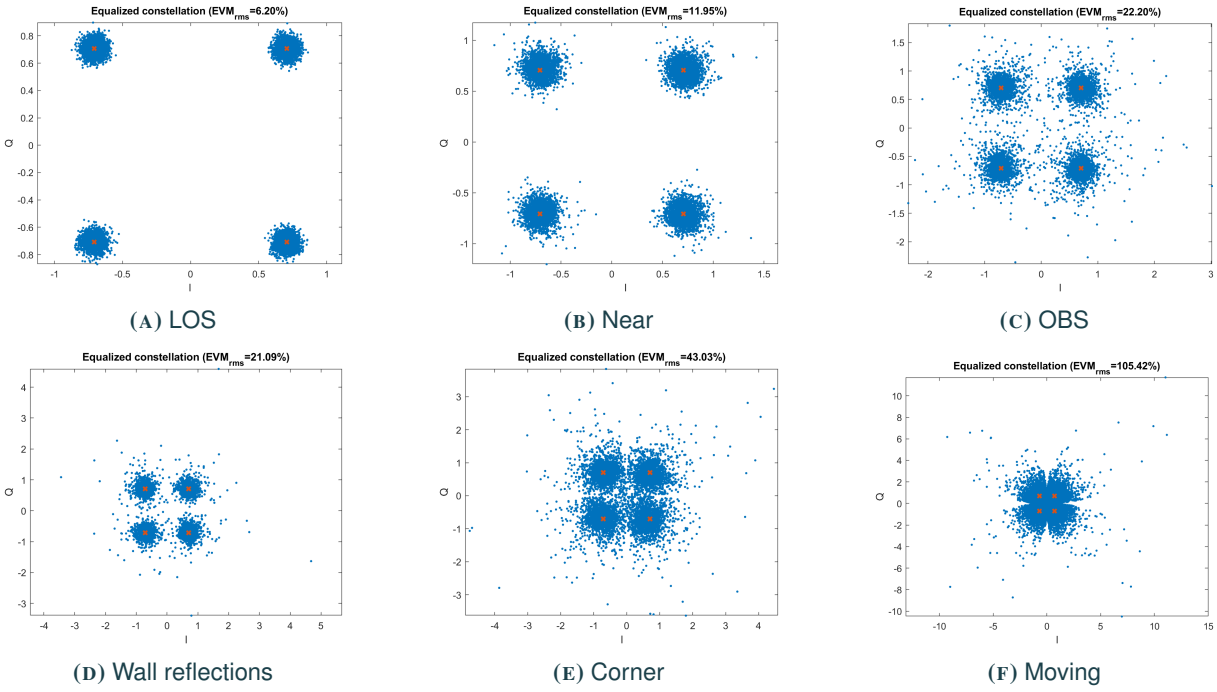


(A) LOS (B) Near (C) OBS

(D) Wall reflections (E) Corner (F) Moving

FIGURE 26

Normalized channel magnitude drift over time. Heatmaps show $20\log_{10}\left|\hat{H}(f,s)/\hat{H}(f,1)\right|$. Values close to 0 dB indicate a time-invariant channel; larger deviations indicate time variation or instability of the effective estimate. The moving scenario exhibits the strongest drift, consistent with its poor constellation quality and high EVM.



(A) LOS (B) Near (C) OBS

(D) Wall reflections (E) Corner (F) Moving

FIGURE 27

Equalized QPSK constellations and $\mathrm{EVM}_{\mathrm{rms}}$. Scatter plots show equalized data symbols after one-tap frequency-domain equalization. Higher EVM reflects residual distortion due to (i) frequency-selective fading (noise enhancement in deep notches of $|\hat{H}_0(f)|$) and (ii) time variation (non-zero drift in Fig. 26). LOS yields tight clusters, while motion collapses the constellation due to rapid channel changes within the frame.

7 ACKNOWLEDGMENTS

We acknowledge Marko Stojanovic and Sofija Orlovic from Group 21 for the helpful discussions and valuable suggestions regarding the model design and possible experiments.

8 APPENDIX

8.1 POWER DELAY PROFILE MEASUREMENT

The power delay profile (PDP) is obtained from the estimated channel impulse response in the time domain. For each received OFDM symbol we first estimate the frequency-domain channel on all active subcarriers,

$$\hat{H}[k] = \frac{Y[k]}{X[k]}, \qquad k = 0, \ldots, N_{\text{car}} - 1,$$

where $X[k]$ and $Y[k]$ denote the transmitted and received QPSK symbols on subcarrier k.

We then compute the corresponding discrete-time channel impulse response by applying an inverse DFT:

$$\hat{h}[n] = \text{IDFT}\{\hat{H}[k]\}, \qquad n = 0, \ldots, N_{\text{car}} - 1.$$

The (instantaneous) PDP for that OFDM symbol is defined as the squared magnitude of the impulse response,

$$P[n] = |\hat{h}[n]|^2,$$

which represents the received power as a function of delay index n (or delay $\tau = nT_s$ with sampling period $T_s = 1/f_s$). In practice, when plotting the PDP we only show the first N_{CP} taps, as these are the ones protected by the cyclic prefix.

To obtain a more stable estimate, we average the PDP over several OFDM symbols and, when needed, over multiple independent recordings:

$$\bar{P}[n] = \frac{1}{N_{\text{sym}}} \sum_{i=1}^{N_{\text{sym}}} |\hat{h}_i[n]|^2.$$

The resulting average PDP $\bar{P}[n]$ is what we plot and compare across different channel scenarios.

8.2 PEAK-TO-AVERAGE POWER RATIO (PAPR)

A well-known drawback of OFDM is its high Peak-to-Average Power Ratio (PAPR). Let $x[n]$ denote the discrete-time baseband OFDM waveform of length N samples. The PAPR is defined as

$$\text{PAPR}(x) = \frac{\max\limits_{0 \le n < N} |x[n]|^2}{\frac{1}{N} \sum\limits_{n=0}^{N-1} |x[n]|^2}.$$

In practice, it is often expressed in decibels as

$$\text{PAPR}_{\text{dB}} = 10 \log_{10}\big(\text{PAPR}(x)\big).$$

In OFDM, many independently modulated subcarriers are added together in the time domain. When their phases align constructively, very large peaks can occur, while the average power remains relatively moderate. This leads to substantially larger PAPR values compared to single-carrier systems.

High PAPR is problematic for practical transmitters:

- **Power amplifier efficiency:** To avoid nonlinear distortion or clipping at the DAC and the power amplifier, the signal must be backed off so that even the rare peaks remain within the linear region. This reduces the effective transmit power and thus the SNR at the receiver.
- **Nonlinear distortion and BER:** If the signal is not sufficiently backed off, the largest peaks are clipped, introducing nonlinear distortion.
- **Audio implementation issues:** In an acoustic OFDM system, excessive PAPR can drive the loudspeaker into a nonlinear regime or saturation, causing audible distortion and additional errors at the receiver.

For these reasons, it is important not only to report the BER performance of the system, but also to monitor the PAPR of the transmitted waveform and, when necessary, to employ PAPR reduction techniques (e.g., phase randomization, interleaving or data whitening) that lower the PAPR without changing the nominal spectral efficiency. In Section 6.3 we will examine in more detail some of these techniques for the image-based OFDM waveform.

8.3 LEAST-SQUARES CHANNEL ESTIMATION

We briefly present least-squares (LS) channel estimation and show how it relates to our channel estimation method. The key idea of LS channel estimation is to estimate the complete channel impulse response within each OFDM symbol, which is possible as long as the actual channel length is shorter than the OFDM symbol length. The LS estimate of the channel impulse response is given by

$$\hat{h} = (T \cdot F_{L\times L})^{\dagger} y_L = (F_{L\times L}^H T^H T F_{L\times L})^{-1} F_{L\times L}^H T^H y_L,$$

where:

- T is a diagonal matrix whose diagonal contains the known training symbols,
- $F_{L\times L}$ is the truncated DFT matrix of size $L \times L$ (with rows/columns removed so that its size matches the assumed channel length L),
- y_L is the received training symbol after selecting only the positions where training symbols were transmitted.

The resulting vector $\hat{h}$ has length L and represents the estimated channel impulse response. To match the full OFDM symbol length N, we zero-pad $\hat{h}$ to length N and compute its FFT:

$$\widehat{H}[m] = \text{FFT}\{\hat{h}_{\text{padded}}\},$$

yielding an estimate of the channel frequency response across all subcarriers. This frequency response is then used to equalize the subsequent OFDM symbols in the frame.

In our actual implementation, we use a slightly simplified version of this idea. From a single known OFDM training symbol we first obtain a per-subcarrier estimate

$$H_{\text{long}}[m] = \frac{Z_{\text{known}}[m]}{A_{\text{known}}[m]},$$

which is precisely the LS solution for each subcarrier of that symbol. We then convert this frequency-domain estimate to the time domain, truncate the impulse response to its significant taps, zero-pad it back to length N and take an FFT again to obtain a smoothed frequency response $\widehat{H}[m]$. For the following data symbols we refine this initial LS estimate with a decision-directed tracker (updating the channel from detected QPSK symbols). Conceptually, this procedure can be viewed as a practical approximation of the full matrix LS scheme: we still start from an LS estimate based on a training symbol, but we use a per-tone formulation and recursive updates rather than explicitly forming the $(TF_{L\times L})^{\dagger}$ matrix for every symbol.

8.4 SETUPS FOR IMAGE TRANSMISSION AND MOVING-MICROPHONE EXPERIMENTS



(A) Baseline LOS setup used in Task 5, static audio conditions.



(B) Image transmission setup in indoor LOS conditions (no obstacle).



(C) Image transmission setup with an obstacle between loudspeaker and microphone.



(D) Audio transmission setup with moving receiver (small left–right hand movements).

FIGURE 28
Overview of the experimental setups used for image transmission and the moving-microphone experiment.

8.5 Error Vector Magnitude (EVM)

Error Vector Magnitude (EVM) quantifies the distance between the equalized received constellation points and the ideal constellation points. Let s_i denote the ideal (reference) QPSK symbols and $\hat{s}_i$ the corresponding equalized received symbols. The error vectors are

$$e_i = \hat{s}_i - s_i. \tag{3}$$

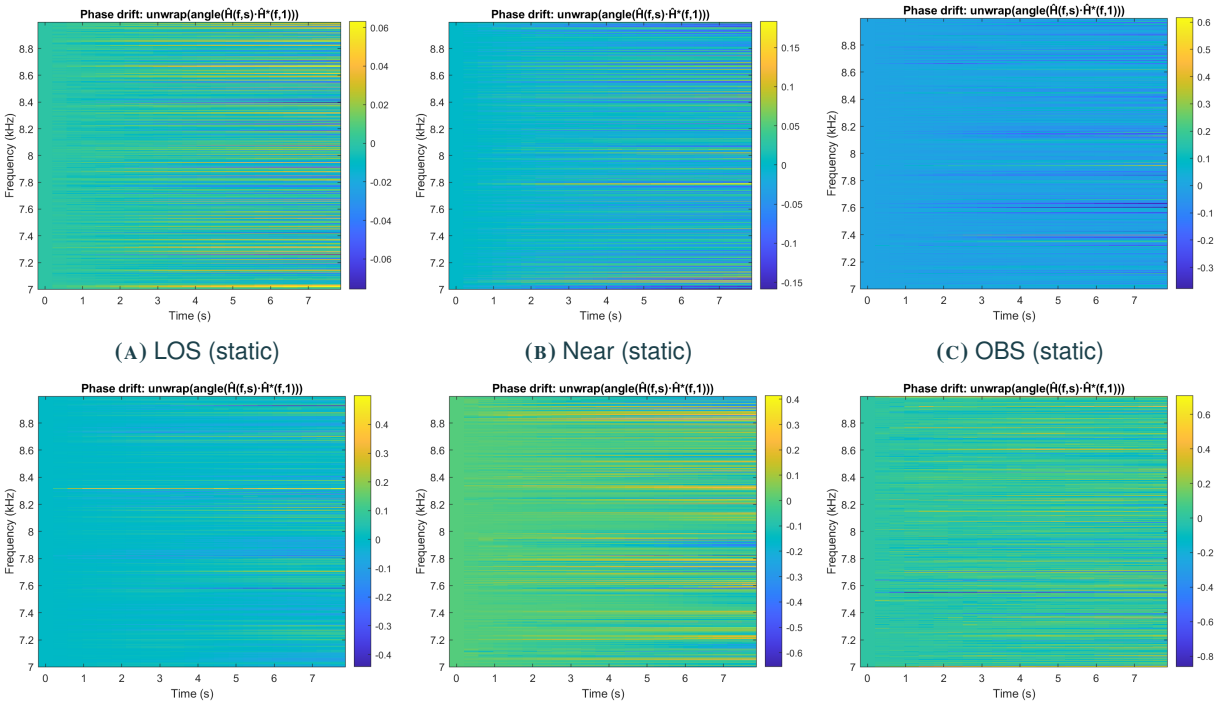
The RMS EVM is defined as

$$\mathrm{EVM}_{\mathrm{rms}}(\%) = 100\sqrt{\frac{\mathbb{E}\left[|e_i|^2\right]}{\mathbb{E}\left[|s_i|^2\right]}}, \tag{4}$$

where $\mathbb{E}[\cdot]$ denotes averaging over symbols. Lower EVM indicates that the received points cluster tightly around the ideal constellation (higher link quality), while higher EVM indicates stronger residual distortion and/or noise.

Implementation note (decision-directed reference). In our implementation, the reference symbols s_i are obtained by mapping each equalized point $\hat{s}_i$ to the nearest ideal QPSK constellation point (minimum-distance decision), rather than using the true transmitted symbol sequence as ground truth. As a result, the reported EVM is an optimistic (lower-bound) estimate of the actual distortion: when symbol decisions are incorrect (e.g., at high BER), using the nearest-point reference can underestimate the error relative to an EVM computed with respect to the true transmitted symbols.

8.6 Phase drift maps



(a) LOS (static) (b) Near (static) (c) OBS (static)

(d) Wall reflections (static) (e) Corner (static) (f) Moving

Figure 29

Per-subcarrier phase drift over time, computed as $\Delta\phi(f,s) = \mathrm{unwrap}\big(\angle(\hat{H}(f,s)\hat{H}^*(f,1))\big)$. Values close to 0 indicate a time-invariant phase response relative to the first OFDM symbol. Static scenarios exhibit small drift; stronger multipath (OBS/wall/corner) shows larger tone-dependent fluctuations, typically on subcarriers affected by deep fades (lower SNR). Motion increases phase instability and frequency-dependent variations, consistent with the observed EVM degradation.

8.7 Hardware Components

The main hardware components of the setup are:

- A laptop running MATLAB, used to generate the OFDM baseband signal, play the audio waveform and record the received signal.
- The internal sound card configured at a sampling frequency of $f_s = 48$ kHz and a bit depth of 16 bits.
- A pair of loudspeakers acting as the *transmitter*.
- A measurement microphone acting as the *receiver*.

8.8 ANDREW VITERBI

Andrew J. Viterbi (1935–) is an Italian-American electrical engineer and co-founder of Qualcomm, best known for the Viterbi algorithm (published in 1967), a landmark method for maximum-likelihood sequence decoding in convolutional codes that became fundamental in digital communications.

In his honor, here we transmit his image using his reknown algorithm. The original and reconstructed images for the LOS experiment using whitening and Viterbi–Viterbi phase tracking are shown in Fig. 30. The resulting performance was $\mathrm{BER} = 0\%$ and $\mathrm{PAPR} = 14.89\ \mathrm{dB}$.



(A) Original image.

(B) Reconstruction (whitening + Viterbi–Viterbi).

FIGURE 30
Image transmission over the audio OFDM link in LOS conditions using whitening to reduce PAPR and Viterbi–Viterbi phase tracking. Achieved performance: $\mathrm{BER} = 0\%$ and $\mathrm{PAPR} = 14.89\ \mathrm{dB}$.